# Project Presentation

Course Report for Reliability and Maintainability Engineering

电子科技大学

University of Electronic Science and Technology of China

# 研究对象与数据介绍

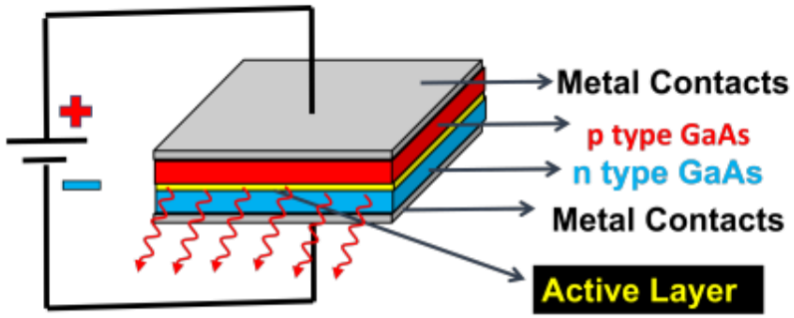
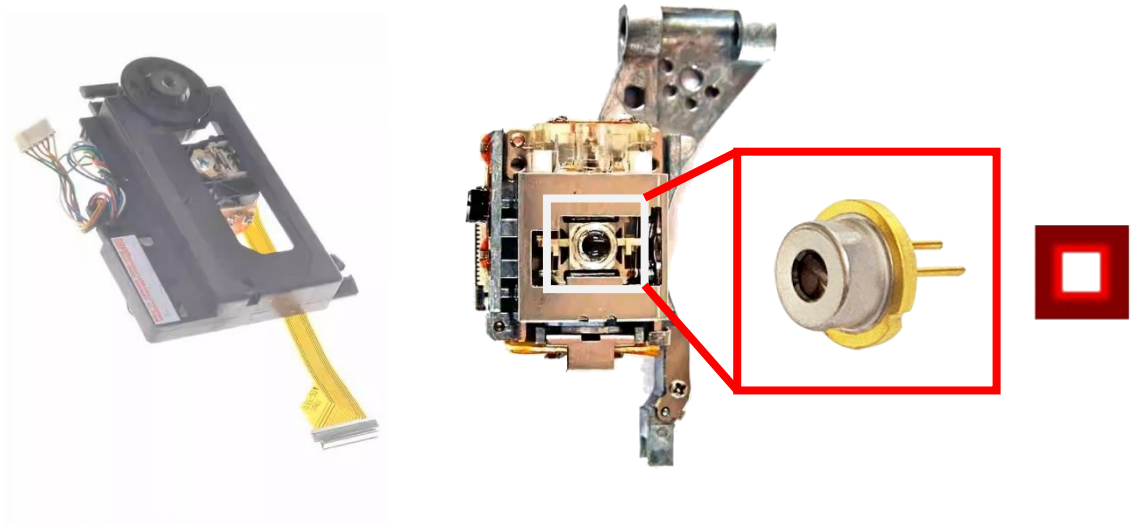
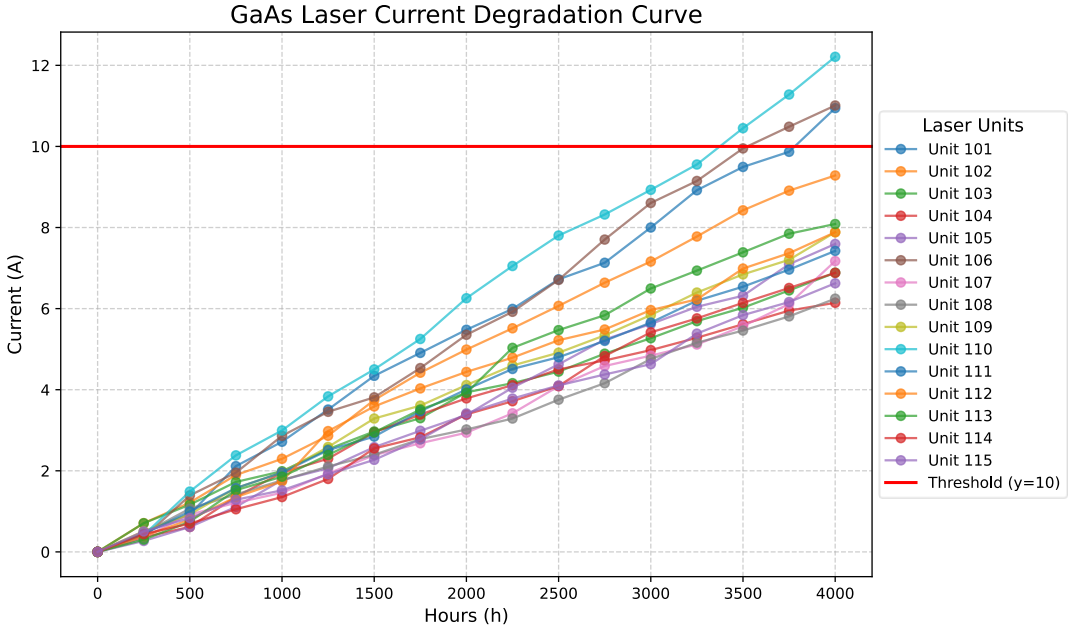



Figure 1: Semiconductor Laser Construction Diagram (GaAs p-n junction)



GaAs激光器

| 编号 | 电流 | 单元 | 小时数 | 批次 |
|---|---|---|---|---|
| 1 | 0 | 101 | 0 | 1 |
| 2 | 0.4741 | 101 | 250 | 1 |
| … | … | … | … | … |
| 6 | 3.511 | 101 | 1250 | 1 |
| 7 | 4.3415 | 101 | 1500 | 1 |
| … | … | … | … | … |
| 16 | 9.8675 | 101 | 3750 | 1 |
| 17 | 10.9446 | 101 | 4000 | 1 |
| … | … | … | … | … |




重复测量加速退化试验

# 退化路径样条拟合——参数估计方法

## The Exponential Degradation Spline Model:

$$X(t) = c \cdot \exp[d \cdot t]$$

### Least Squares Method Parameter Estimation

$$\begin{aligned} loss(c,d) &= \sum_{j=1}^{m}\left[\ln x_j - \ln x_j\right]^2 \\ &= \sum_{j=1}^{m}\left[\ln x_j - (dt_j + \ln c)\right]^2 \\ &= \sum_{j=1}^{m}\left[(\ln x_i)^2 - 2dt_j \ln x_j - 2\ln c \ln x_j + d^2 t_j^2 + 2dt_j \ln c + (\ln c)^2\right] \end{aligned}$$

$$\begin{cases} \dfrac{\partial loss}{\partial d} = \sum_{j=1}^{m}\left[-2\ln x_j t_j + 2dt_i^2 + 2t_i \ln c\right] \\ \dfrac{\partial loss}{\partial c} = \sum_{j=1}^{m}\left[-2\ln x_j + 2dt_i + 2\ln c\right] \end{cases}$$

$$\begin{cases} \hat{d} = \dfrac{n\sum_{j=1}^{n}\ln x_j t_j - \sum_{j=1}^{m}\ln x_j \sum_{j=1}^{m}\ln t_j}{n\sum_{j=1}^{m}t_i^2 - \left(\sum_{j=1}^{m}t_i\right)^2} \\ \hat{c} = \exp\left(\dfrac{\sum_{j=1}^{m}\ln x_j - \hat{d}\sum_{j=1}^{m}t_i}{n}\right) \end{cases}$$

**Derivation Trick:**

$$n \cdot \bar{x} = \sum_{j=1}^{n} x_j$$

$$n \cdot \bar{t} = \sum_{j=1}^{m} t_j$$

## The Power Degradation Spline Model:

$$X(t) = c \cdot t^d$$

### Least Squares Method Parameter Estimation

$$\begin{aligned} loss(c,d) &= \sum_{j=1}^{m}\left[\ln x_j - \ln x_j\right]^2 \\ &= \sum_{j=1}^{m}\left[\ln x_j - (d\ln t_j + \ln c)\right]^2 \\ &= \sum_{j=1}^{m}\left[(\ln x_i)^2 - 2d\ln t_j \ln x_j - 2\ln c \ln x_j + d^2(\ln t_j)^2 + 2d\ln t_j \ln c + (\ln c)^2\right] \end{aligned}$$

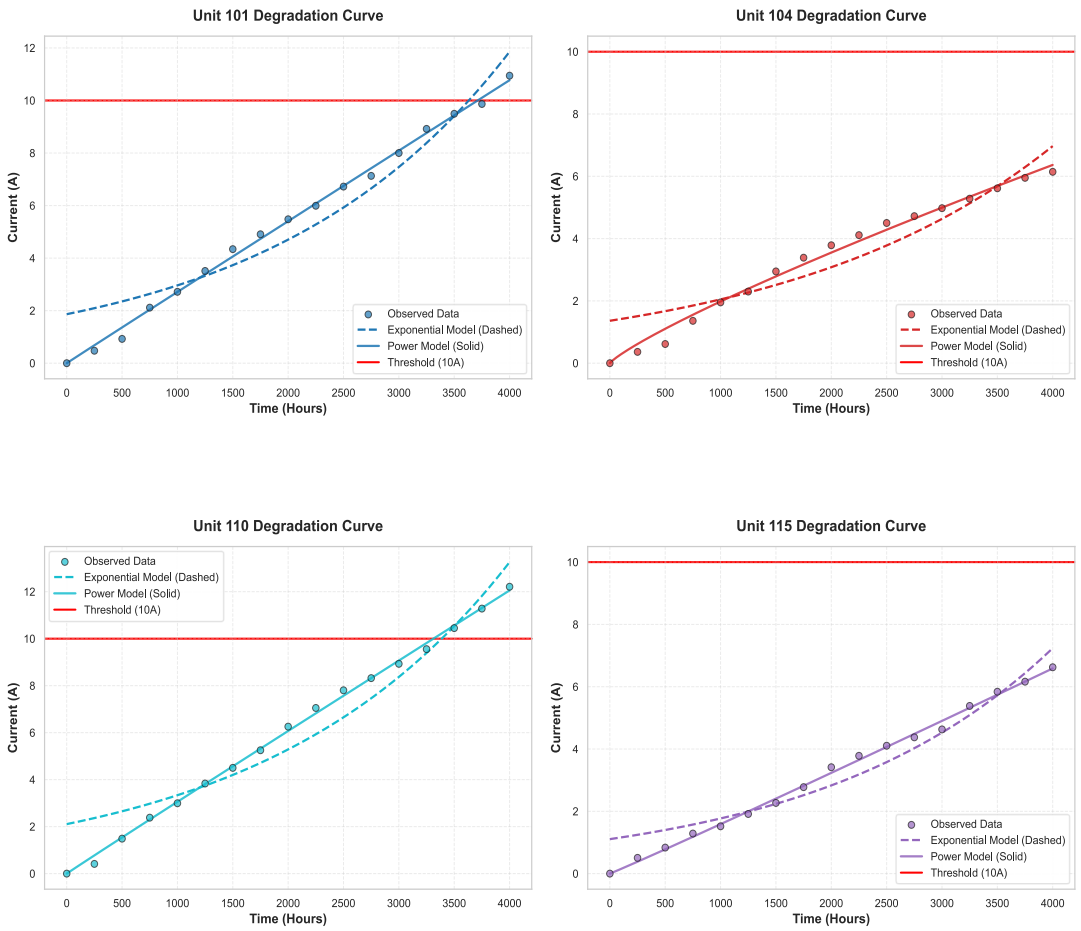
$$\begin{cases} \dfrac{\partial loss}{\partial d} = \sum_{j=1}^{m}\left[-2\ln x_j \ln t_j + 2d(\ln t_j)^2 + 2\ln c \ln t_j\right] \\ \dfrac{\partial loss}{\partial c} = \sum_{j=1}^{m}\left[-2\ln x_j + 2d\ln t_i + 2\ln c\right] \end{cases}$$

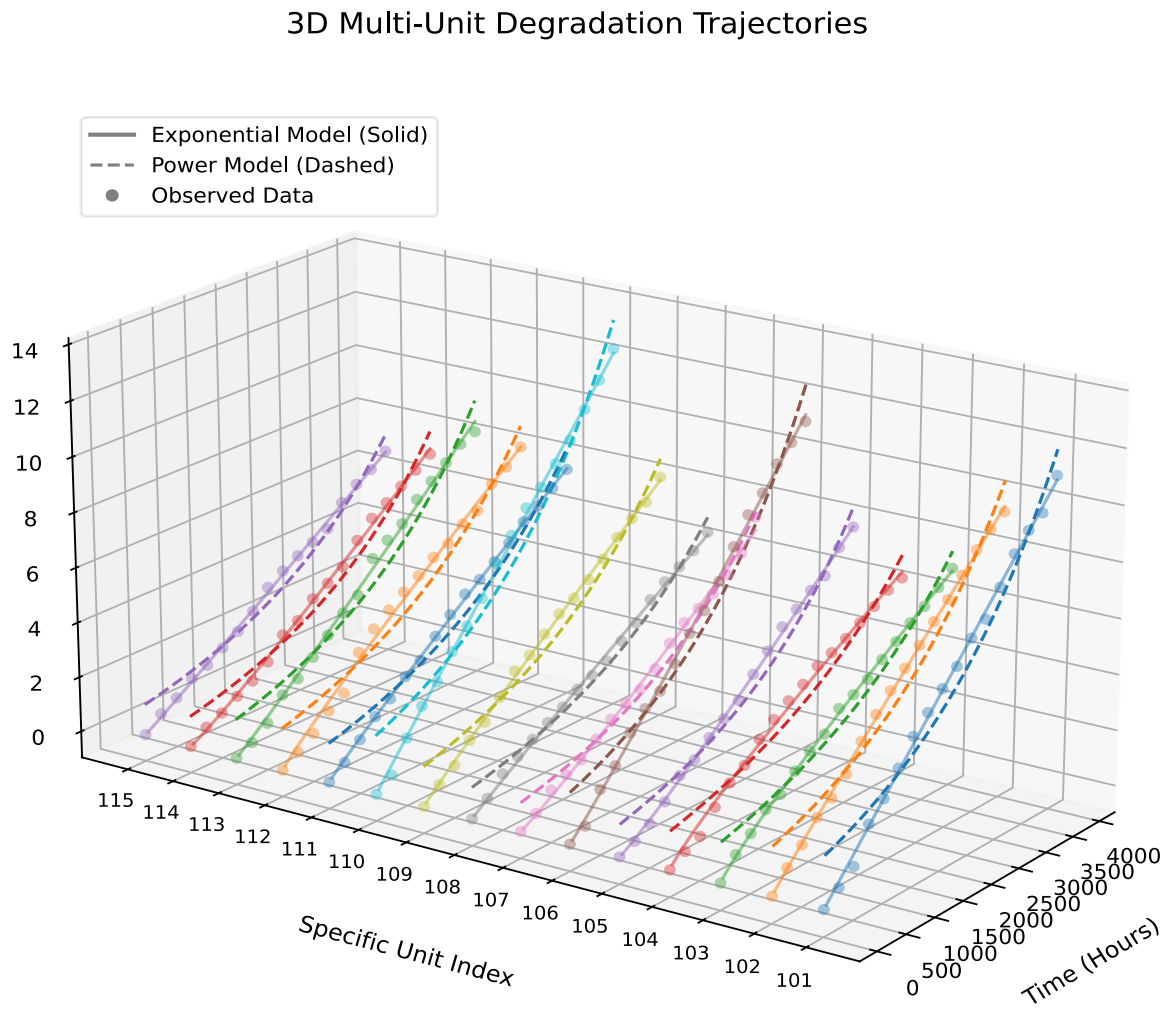
$$\begin{cases} \hat{d} = \dfrac{n\sum_{j=1}^{n}\ln x_j \ln t_j - \sum_{j=1}^{m}\ln x_j \sum_{j=1}^{m}\ln t_j}{n\sum_{j=1}^{m}(\ln t_j)^2 - \left(\sum_{j=1}^{m}\ln t_i\right)^2} \\ \hat{c} = \exp\left(\dfrac{\sum_{j=1}^{m}\ln x_j - \hat{d}\sum_{j=1}^{m}\ln t_j}{n}\right) \end{cases}$$

Another way is using optimal methos

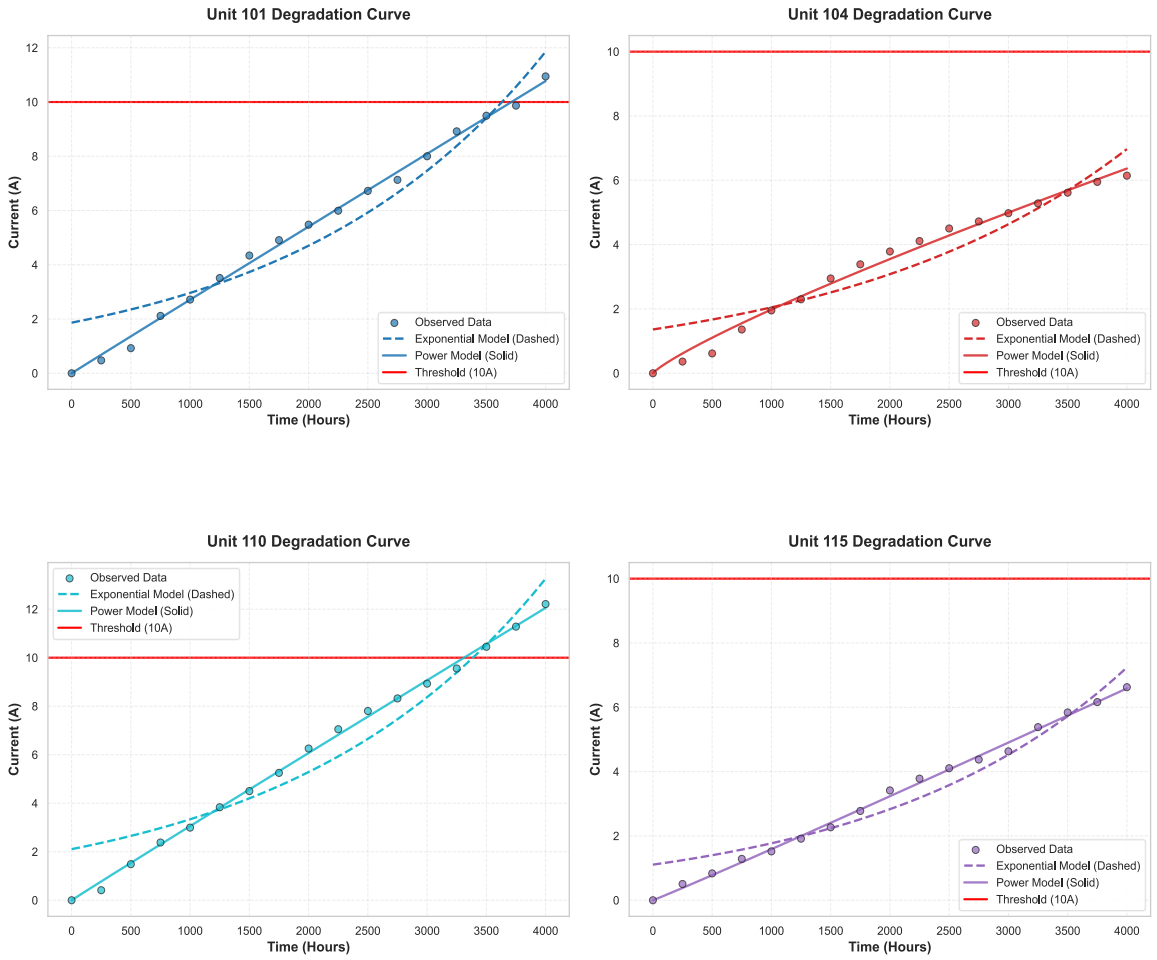
scipy.optimize.curve_fit

# 退化路径样条拟合——拟合路径可视化



# 退化路径样条拟合——失效时间预测




Exponential:

$$c \cdot \exp(d \cdot TTF) = \delta$$

$$d \cdot TTF = \ln\delta - \ln c$$

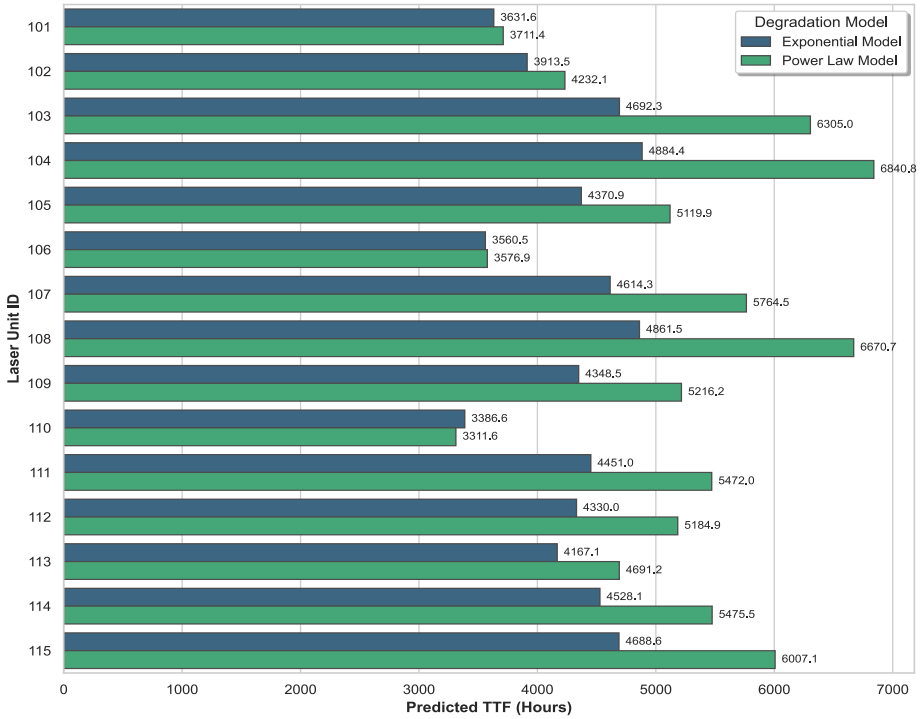
$$TTF = \frac{\ln\delta - \ln c}{d}$$

Power:

$$c \cdot TTF^{d} = \delta$$

$$d \cdot \ln TTF = \ln\delta - \ln c$$

$$TTF = \exp\left[\frac{\ln\delta - \ln c}{d}\right]$$



# 退化路径样条拟合——模型选择

$$MSE = \sum_{j=1}^{m}(y_j - y_j)^2$$

The lower the better

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_{j=1} m(y_j - y_j)^2}{\sum_{j=1} m(y_j - \bar{y})^2}$$

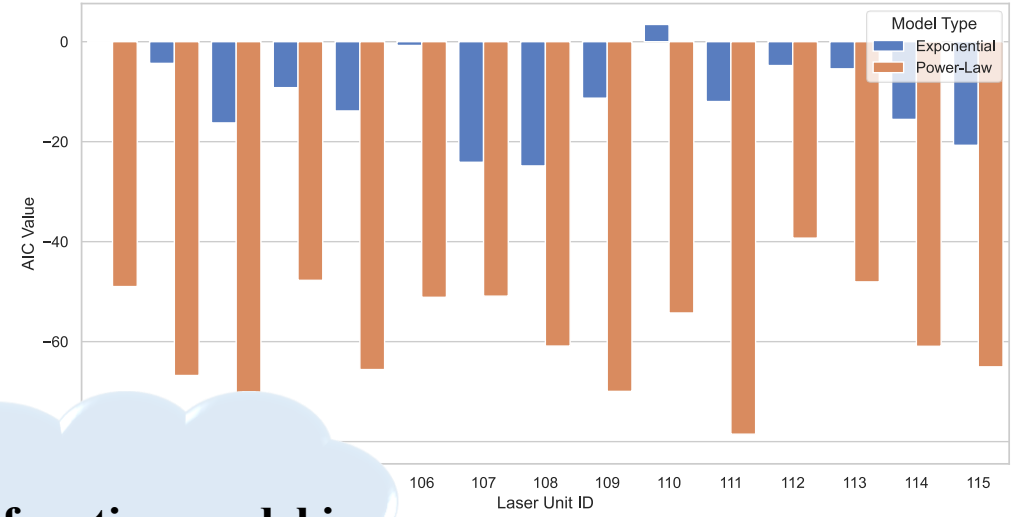
The closer to 1 the better

$$AIC = -2\ln(\hat{L}) + 2k = n\ln\left(\frac{RSS}{n}\right) + 2k$$

The lower the better

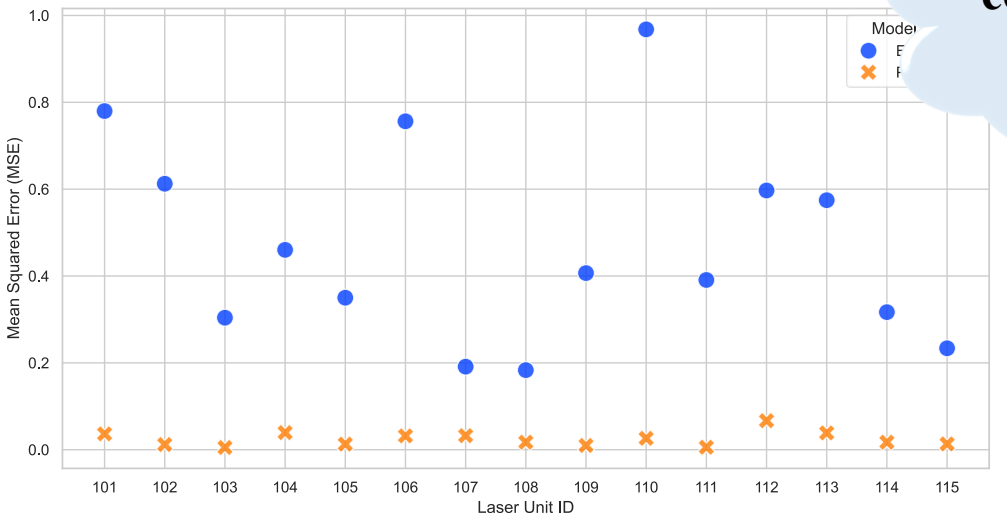
Assume the error follows normal distribution:

$$\epsilon = y_j - y_j \sim N(0, \sigma^2)$$

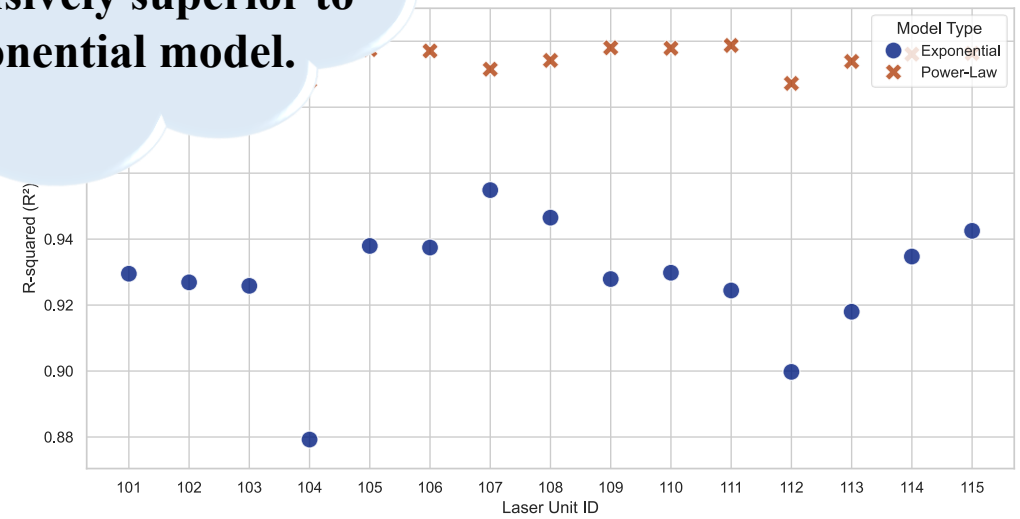



AIC Result

**The power function model is comprehensively superior to the exponential model.**




MSE Result



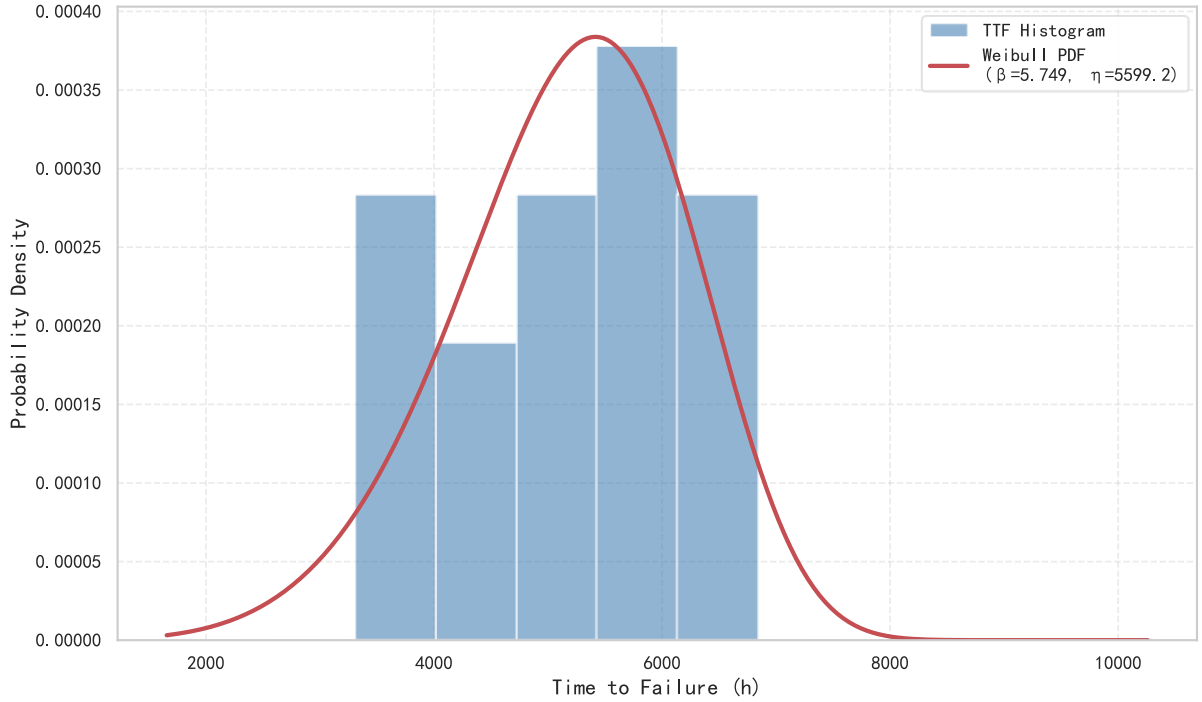

R-squared Result

# 基于威布尔分布的失效时间拟合

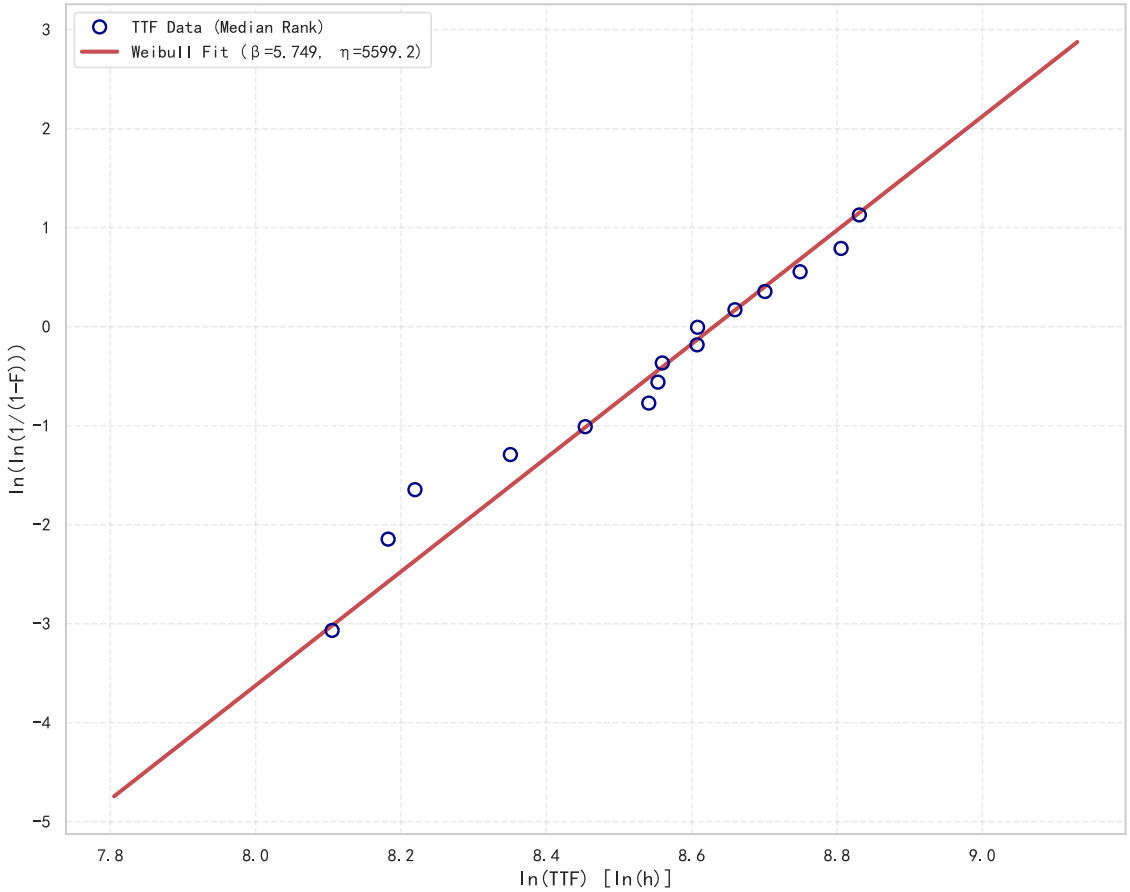
Two-para Weibull Distribution

$$f(t;\beta,\eta)=\frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1}\exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right]$$

scipy.stats.weibull_min.fit(ttf_data, floc=0)

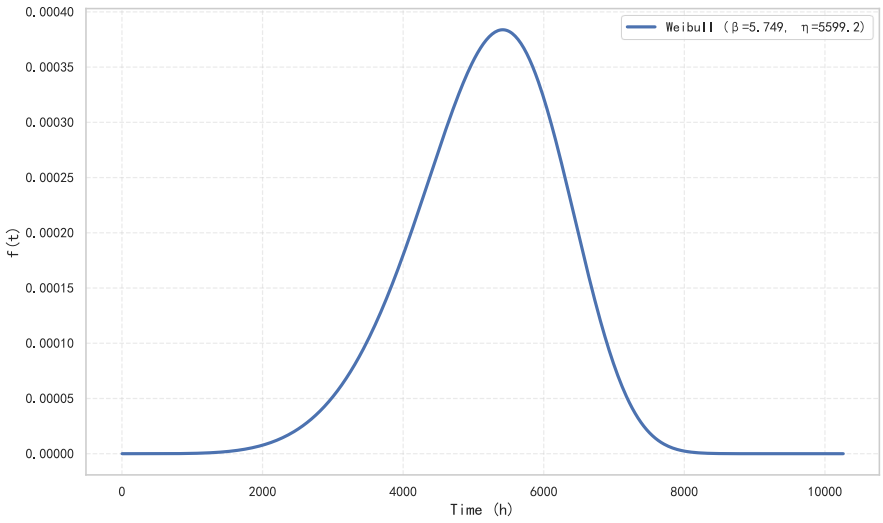



Weibull PDF with Sample Histogram

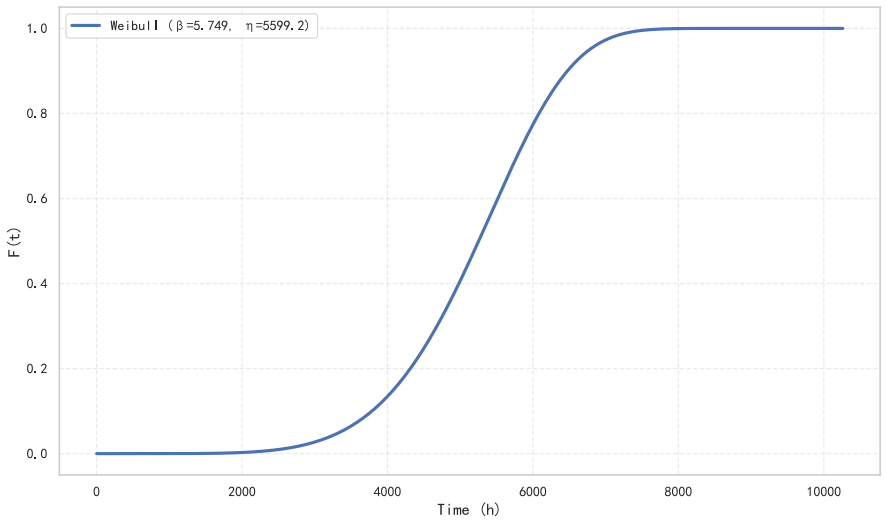



Weibull Probability Plot

# 基于威布尔分布的失效时间分析



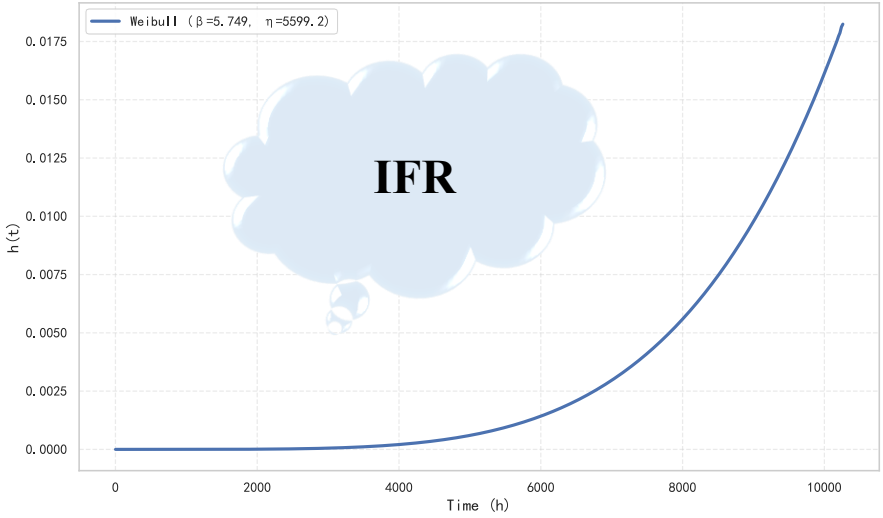

Probability Density Function



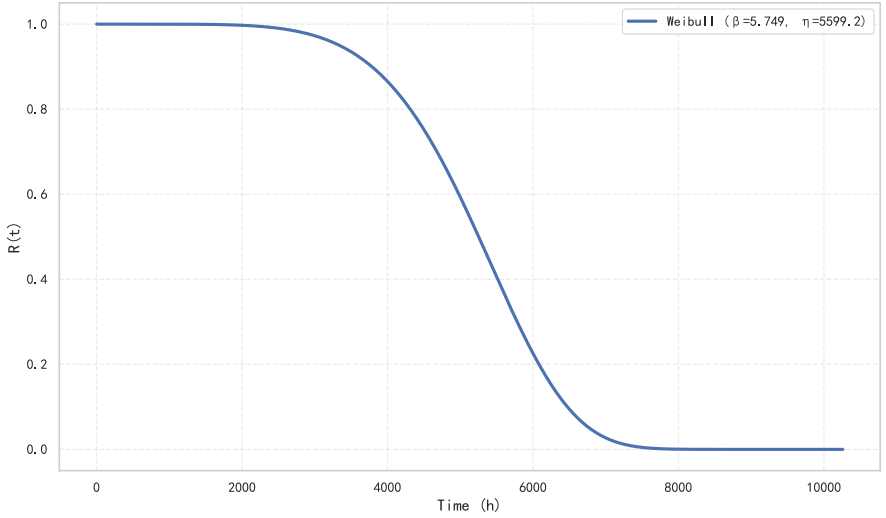

Cumulative Distribution Function

$$f(t)=\frac{5.749}{5599.2}\left(\frac{t}{5599.2}\right)^{4.749}e^{-\left(\frac{t}{5599.2}\right)^{5.749}}$$

$$F(t)=1-e^{-\left(\frac{t}{\eta}\right)^{\beta}}=1-e^{-\left(\frac{t}{5599.2}\right)^{5.749}}$$




Failure Rate/ Hazard Function

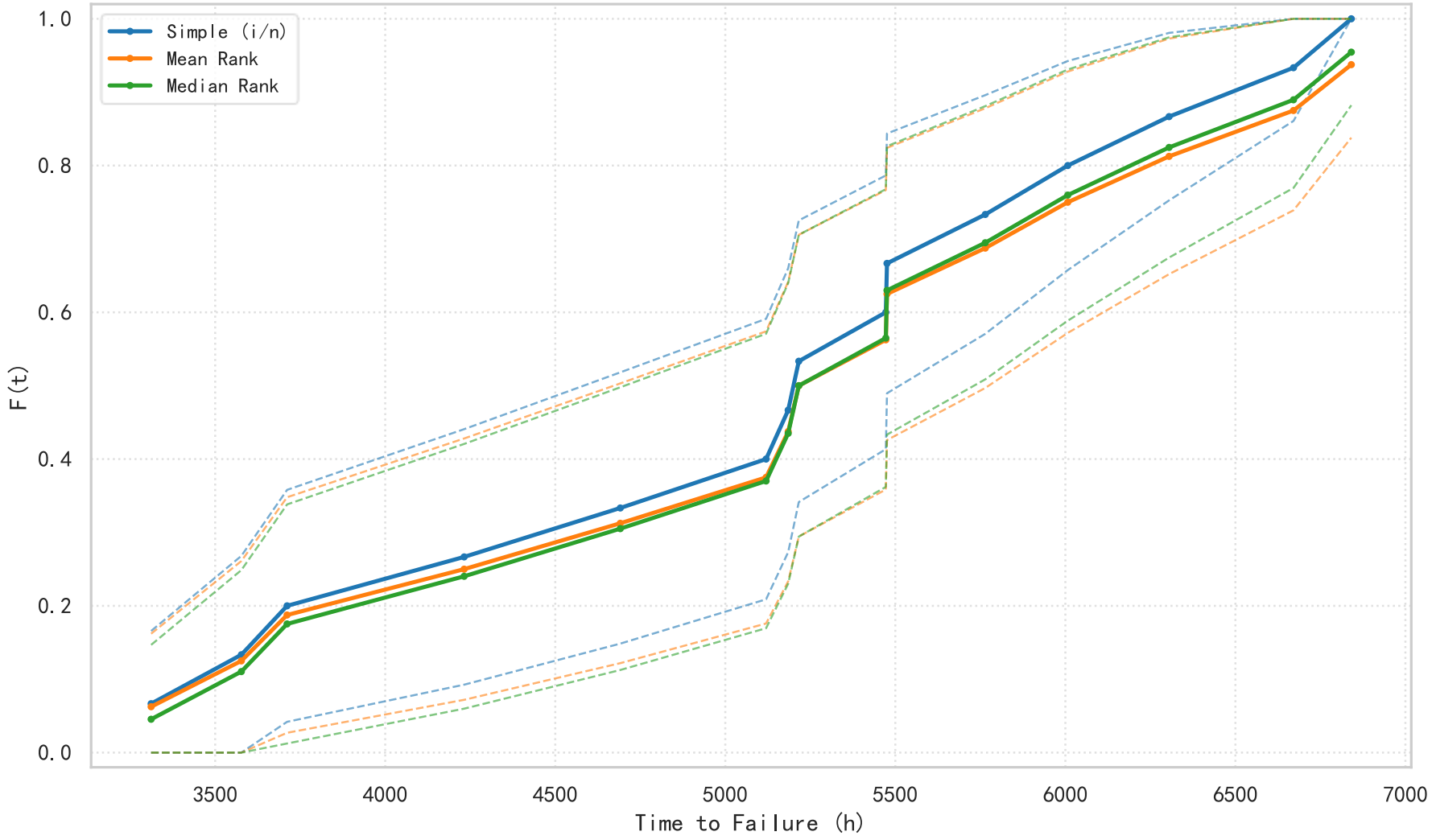



Reliability Function

$$h(t)=\frac{f(t)}{R(t)}=\frac{5.749}{5599.2}\left(\frac{t}{5599.2}\right)^{4.749}$$

$$R(t)=e^{-\left(\frac{t}{\eta}\right)^{\beta}}=e^{-\left(\frac{t}{5599.2}\right)^{5.749}}$$

# 基于经验方法的评估——累计分布函数




Cumulative Distribution Function

Simple Rank

$$F(t_i) = \frac{i}{n}$$

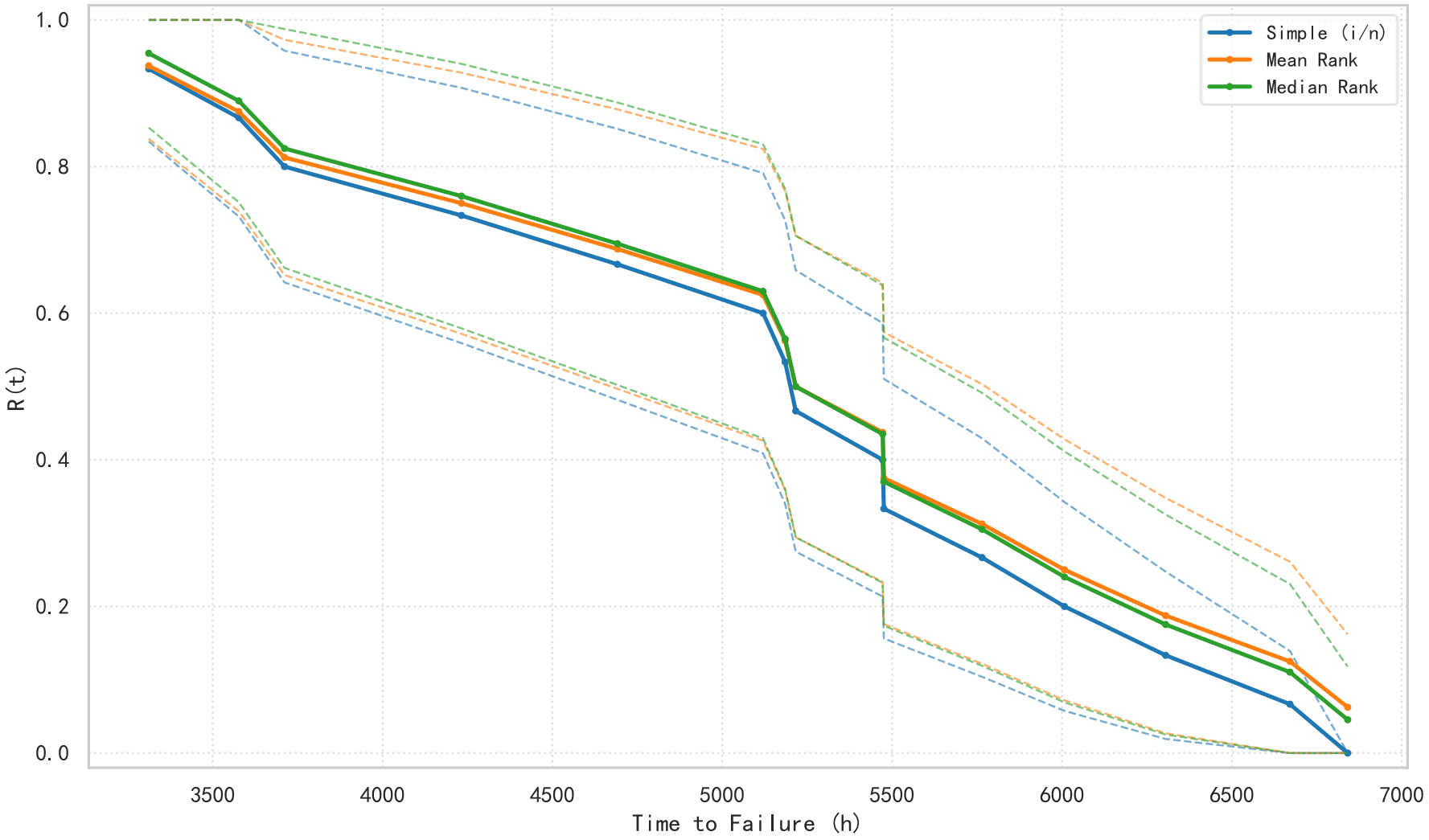
Mean Rank

$$F(t_i) = \frac{i}{n+1}$$

Median Rank

$$F(t_i) = \frac{i-0.3}{n+0.4}$$

# 基于经验方法的评估——可靠性




Reliability Function

Simple Rank

$$R(t_i) = 1 - \frac{i}{n} = \frac{n-i}{n}$$

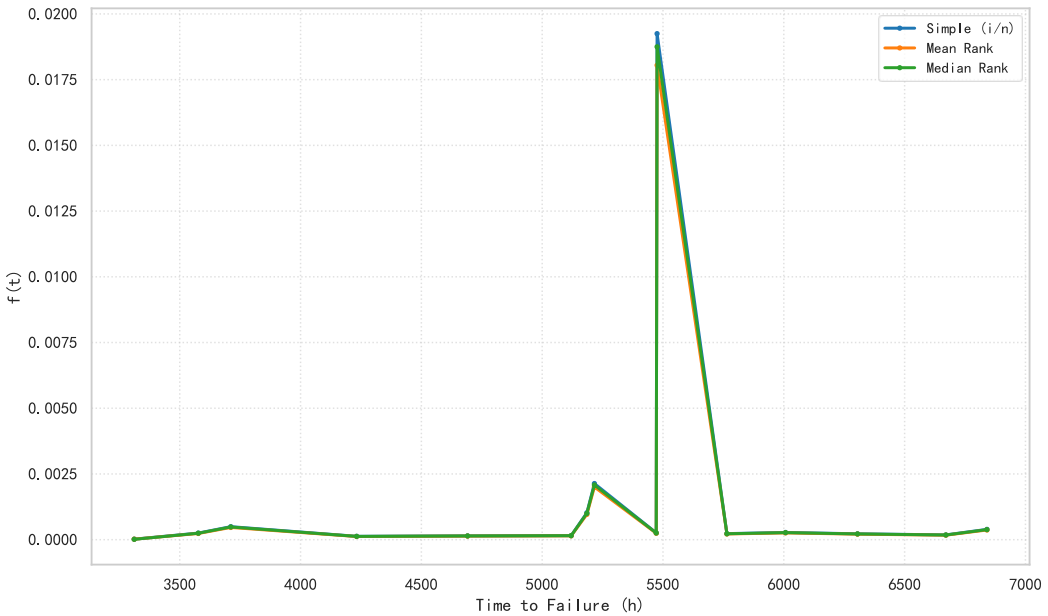
Mean Rank

$$R(t_i) = 1 - \frac{i}{n+1} = \frac{n+1-i}{n+1}$$

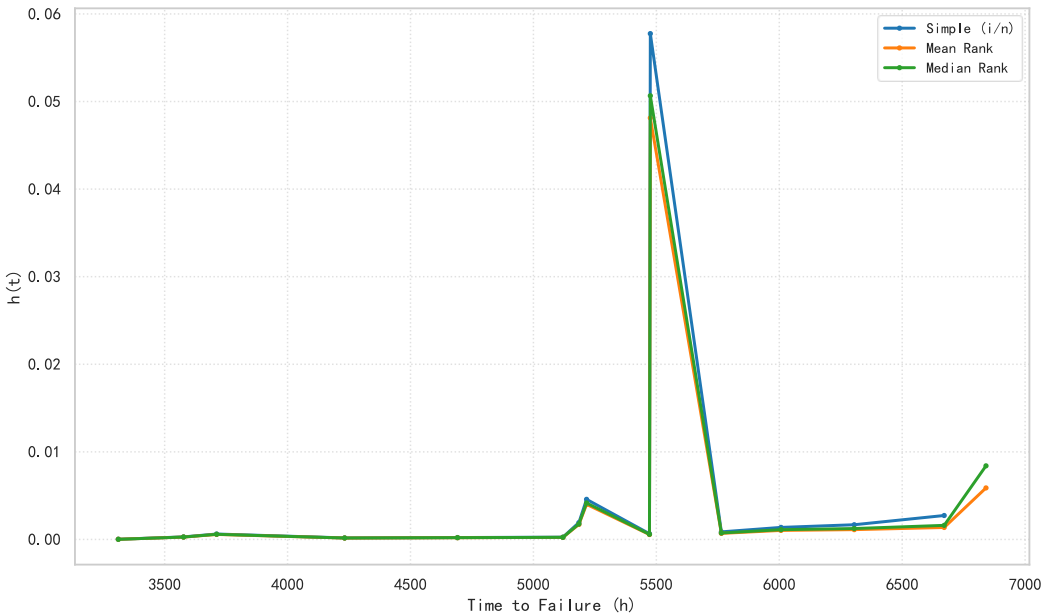
Median Rank

$$R(t_i) = 1 - \frac{i-0.3}{n+0.4} = \frac{n+0.7-i}{n+0.4}$$

# 基于经验方法的评估——PDF & FR




Probability Density Function

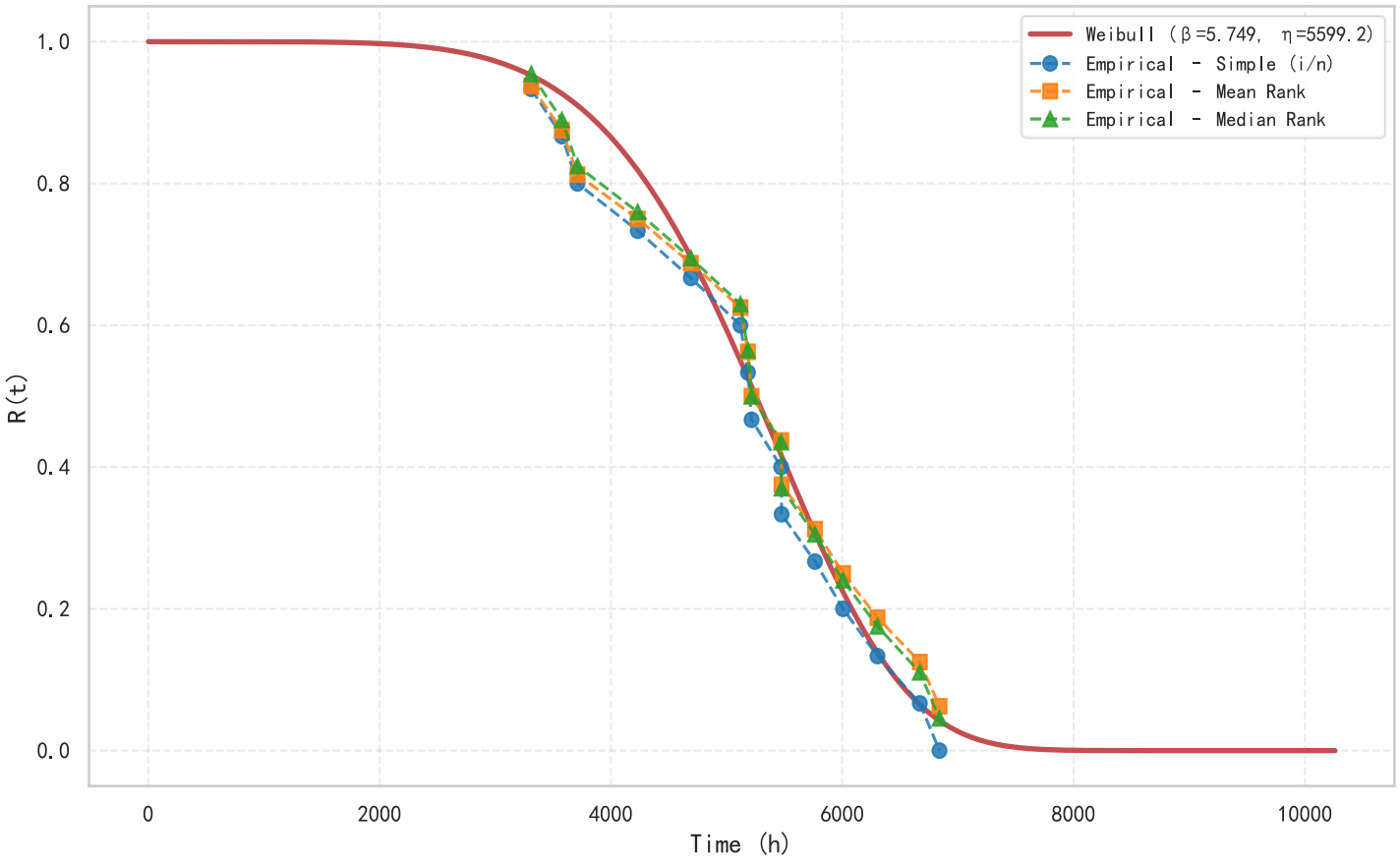



Failure Rate/ Hazard Function

$$f(t_i) = \frac{F(t_i) - F(t_{i-1})}{t_i - t_{i-1}}$$

$$h(t_i) = \frac{f(t_i)}{R(t_i)}$$

# 经验方法与参数方法的对比

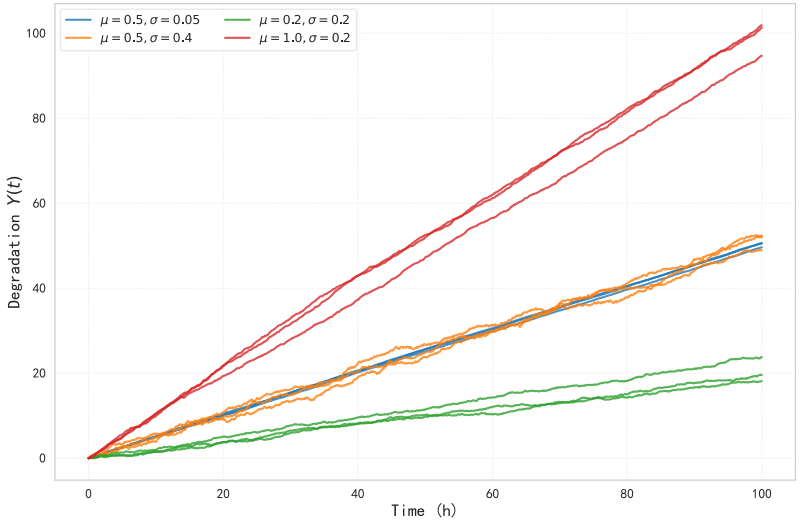



| Method | Median Life (h) |
|---|---|
| Weibull | 5253.364 |
| Simple | 5200.547 |
| Mean Rank | 5216.151 |
| Median Rank | 5216.151 |

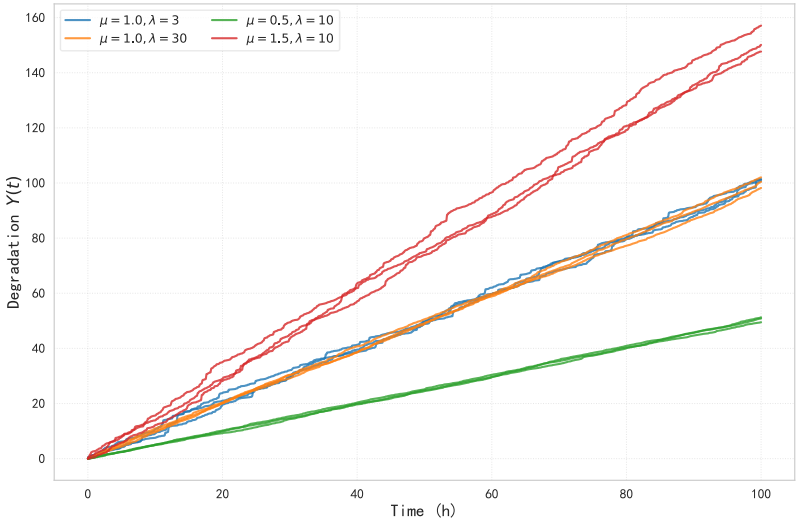
# 基于随机过程的退化建模分析

Degradation-based reliability modeling characterizes **system health** as a **stochastic process**, defining failure as the moment degradation **crosses a critical threshold** to predict reliability and remaining useful life.

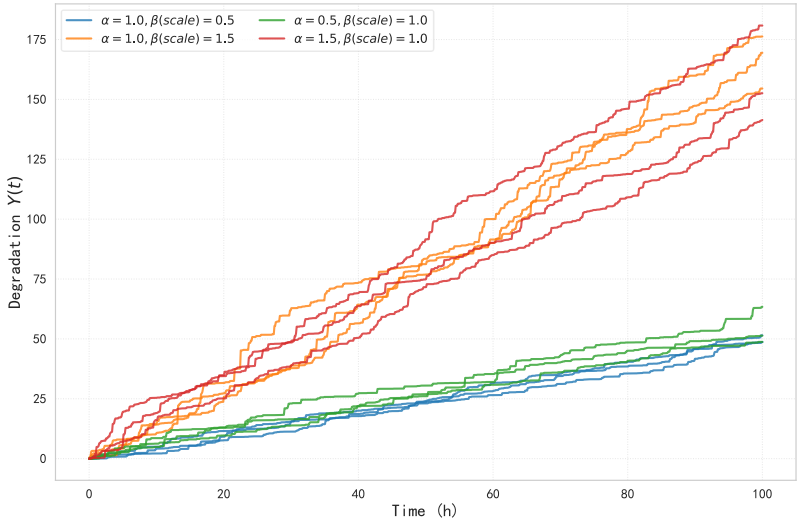
退化可靠性建模将系统性能演变视为随机过程，通过计算退化量首次超越失效阈值的概率，实现对高可靠性产品的寿命评估与余寿预测




Wiener Process




Inverse Gaussian Process




Gamma Process

# 基于Wiener过程的退化建模分析——MLE参数估计

$$X(t) = \mu t + \sigma B(t)$$

The increments are independent and follow normal distribution: $\Delta y_j \sim N(\mu \Delta t_j, \sigma^2 \Delta t_j)$

with probability density function: $$f(\Delta y_j \mid \mu, \sigma^2, \Delta t_j) = \frac{1}{\sqrt{2\pi\sigma^2 \Delta t_j}} \exp\left( -\frac{(\Delta y_j - \mu \Delta t_j)^2}{2\sigma^2 \Delta t_j} \right)$$

thus the likelihood function is: $$L(\mu, \sigma^2) = \prod_{j=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2 \Delta t_j}} \exp\left( -\frac{(\Delta y_j - \mu \Delta t_j)^2}{2\sigma^2 \Delta t_j} \right)$$

log-likelihood: $$l(\mu, \sigma^2) = -\frac{N}{2}\ln(2\pi) - \frac{N}{2}\ln(\sigma^2) - \frac{1}{2}\sum_{j=1}^{N}\ln(\Delta t_j) - \sum_{j=1}^{N}\frac{(\Delta y_j - \mu \Delta t_j)^2}{2\sigma^2 \Delta t_j}$$

Maximizing the log-likelihood function via the first-order necessary conditions:

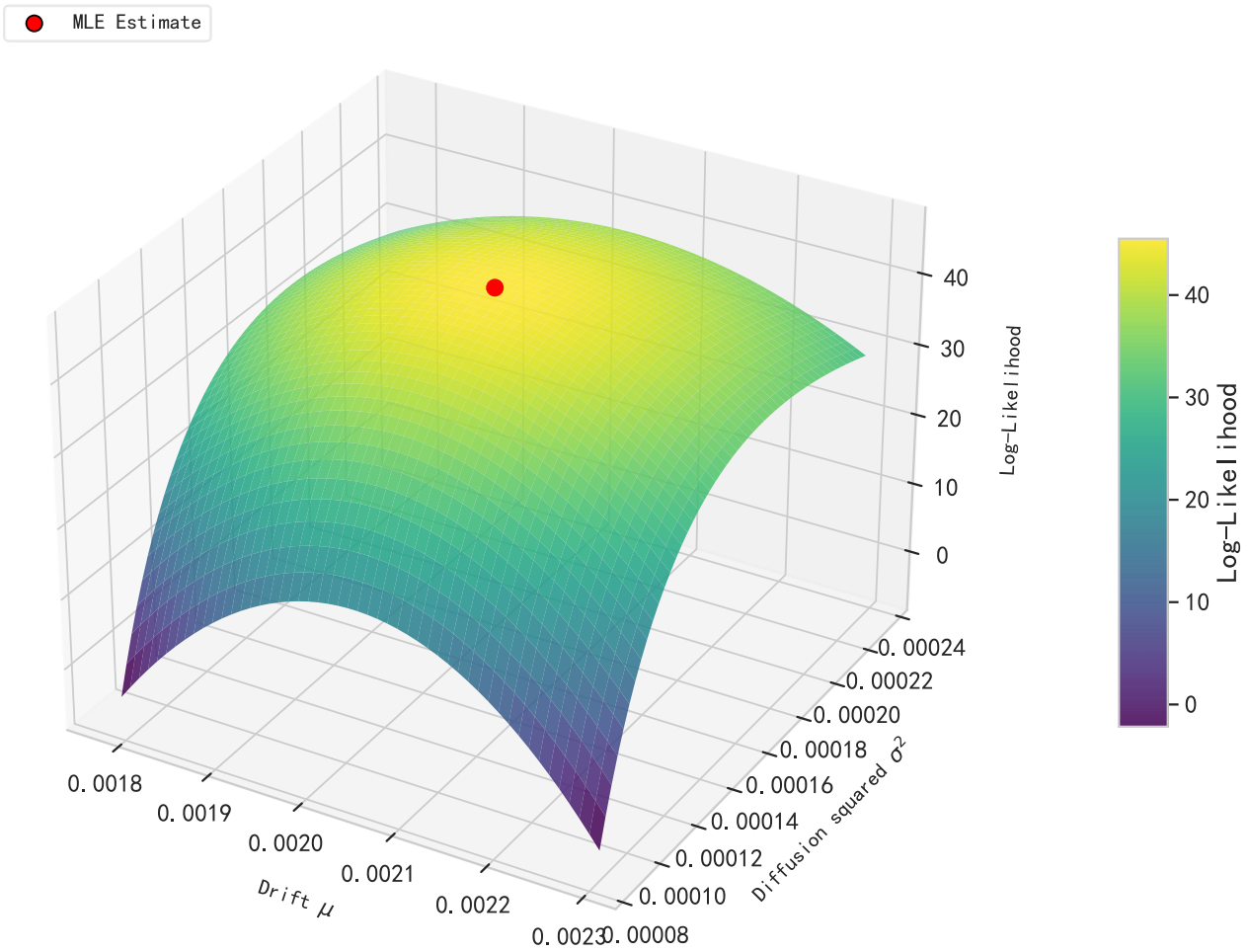
$$\frac{\partial l}{\partial \mu} = \frac{1}{\sigma^2}\sum_{j=1}^{N}\frac{(\Delta y_j - \mu \Delta t_j)\cdot \Delta t_j}{\Delta t_j} = \frac{1}{\sigma^2}\sum_{j=1}^{N}(\Delta y_j - \mu \Delta t_j) = 0$$

$$\frac{\partial l}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{j=1}^{N}\frac{(\Delta y_j - \mu \Delta t_j)^2}{\Delta t_j} = 0$$

$$\Rightarrow \quad \hat{\mu} = \frac{\sum_{j=1}^{N}\Delta y_j}{\sum_{j=1}^{N}\Delta t_j}$$

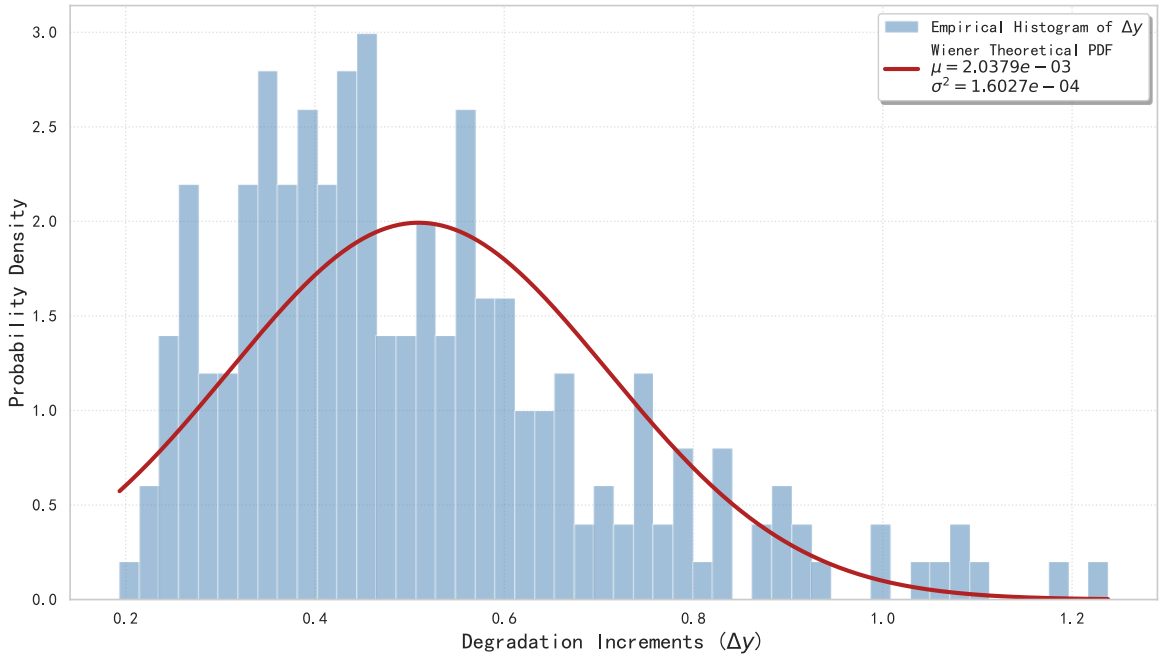
$$\hat{\sigma}^2 = \frac{1}{N}\sum_{j=1}^{N}\frac{(\Delta y_j - \hat{\mu}\Delta t_j)^2}{\Delta t_j}$$

# 基于Wiener过程的退化建模分析——MLE参数估计




Visualization of Log-likelihood Surface

$$L(\theta) = \prod_{i=1}^{N} f(\theta; x_i)$$




Probability Density Function of Degradation Increments

# 基于Wiener过程的退化建模分析——可靠性评估

$$X(t) = \mu t + \sigma B(t) \qquad T = \inf\{t > 0 : X(t) \geq L\}$$

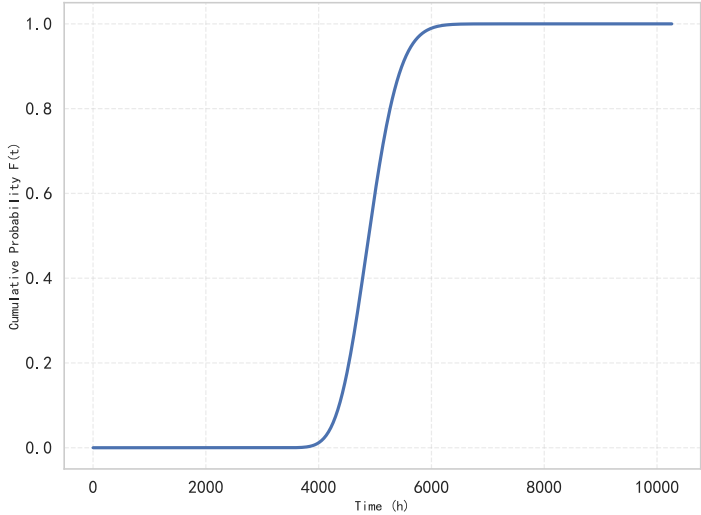
$$F(t) = P(T \leq t) = P\left(\max_{0 \leq s \leq t} X(s) \geq L\right)$$

Fokker-Planck Equation (also known as the Kolmogorov Forward Equation) or Girsanov Change

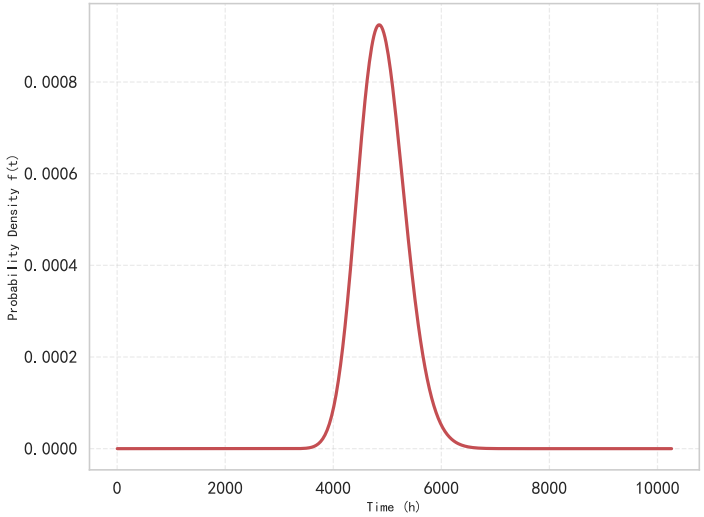
**Inverse Gaussian Distribution**

$$F(t;\mu,\sigma^2,L) = \Phi\left(\frac{\mu t - L}{\sqrt{\sigma^2 t}}\right) + e \qquad \sqrt{\sigma^2 t}\Big)$$

$$f(t) = \frac{L}{\sqrt{2\pi\sigma^2 t^3}} \exp\left[-\frac{(L-\mu t)^2}{2\sigma^2 t}\right] \qquad h(t) = \frac{f(t)}{1 - F(t)}$$

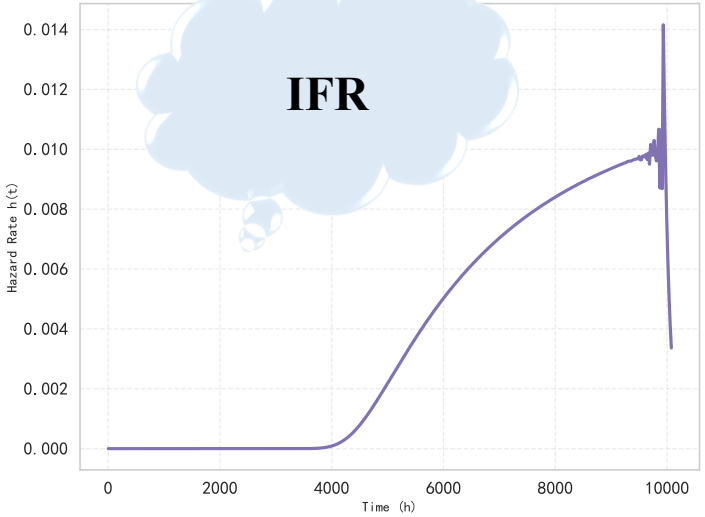



CDF of Failure Time

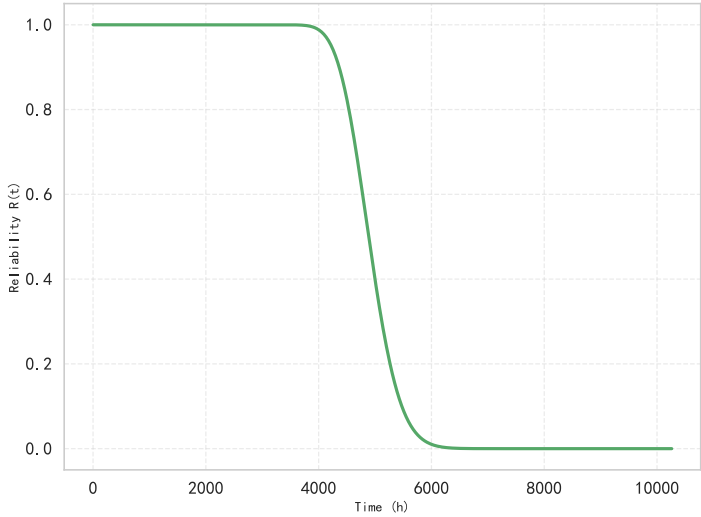



PDF of Failure Time

**IFR**




Failure Rate Function



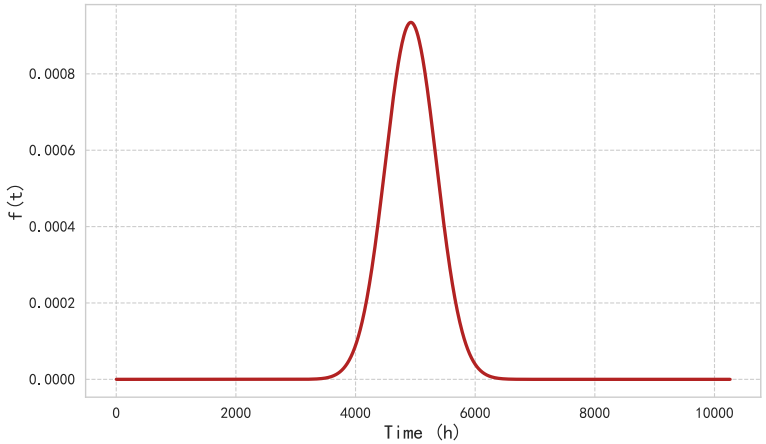

Reliability Function

# 基于逆高斯过程的退化建模分析

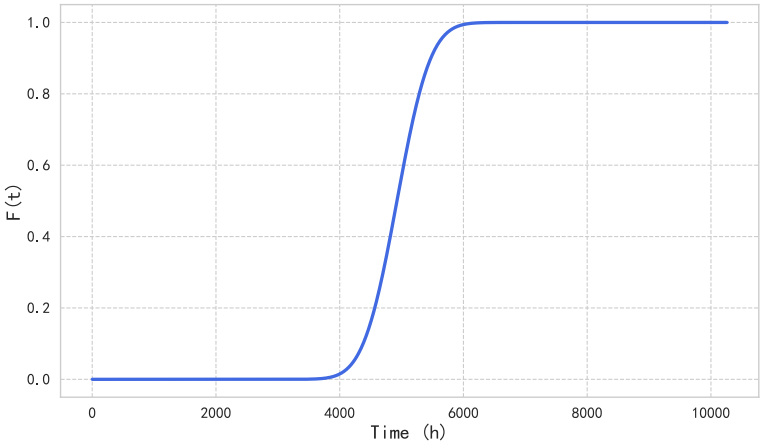
$$\Delta y \sim \mathrm{IG}(a\Delta t, b\Delta t^2)$$

$$f_{\Delta Y}(\Delta y) = \sqrt{\frac{b\Delta t^2}{2\pi\Delta y^3}} \exp\left[-\frac{b\Delta t^2(\Delta y - a\Delta t)^2}{2a^2\Delta t^2\Delta y}\right]$$

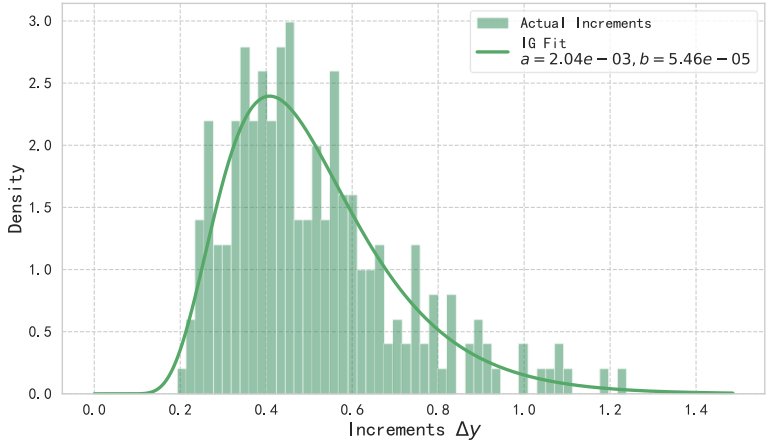
$$F_T(t) = P(Y(t) \geq L)$$
$$= 1 - \mathrm{CDF}_{\mathrm{IG}}(L; at, bt^2)$$



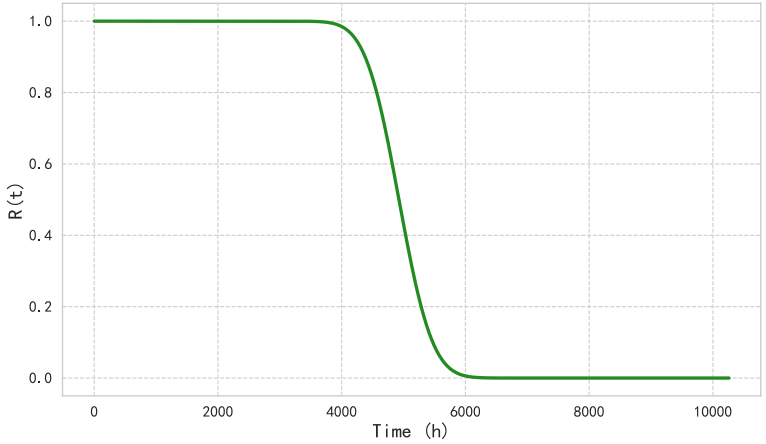

PDF of Failure Time



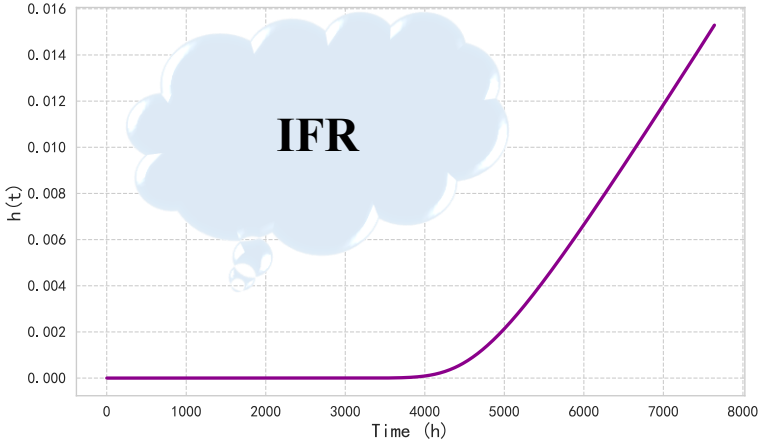

CDF of Failure Time




PDF of Degradation Increments




Reliability Function



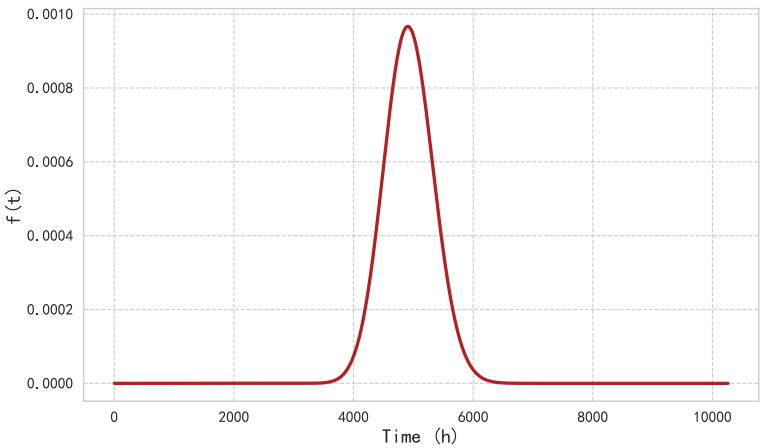

Failure Rate Function

# 基于Gamma过程的退化建模分析

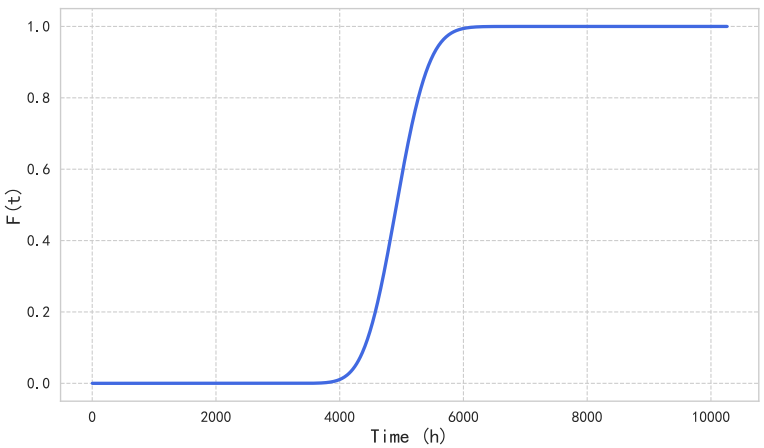
$$\Delta y \sim \mathrm{Gamma}(\alpha\Delta t, \beta)$$

$$f_{\Delta Y}(\Delta y) = \frac{1}{\Gamma(\alpha\Delta t)\beta^{\alpha\Delta t}} \Delta y^{\alpha\Delta t - 1} e^{-\frac{\Delta y}{\beta}}$$

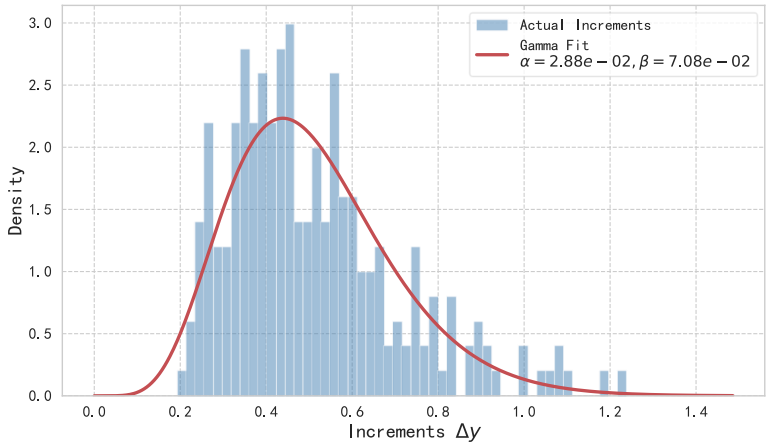
$$\begin{aligned} F_T(t) &= P(T \le t) \\ &= P(Y(t) \ge L) \\ &= 1 - \mathrm{CDF}_{\mathrm{Gamma}}(L; \alpha t, \beta) \end{aligned}$$



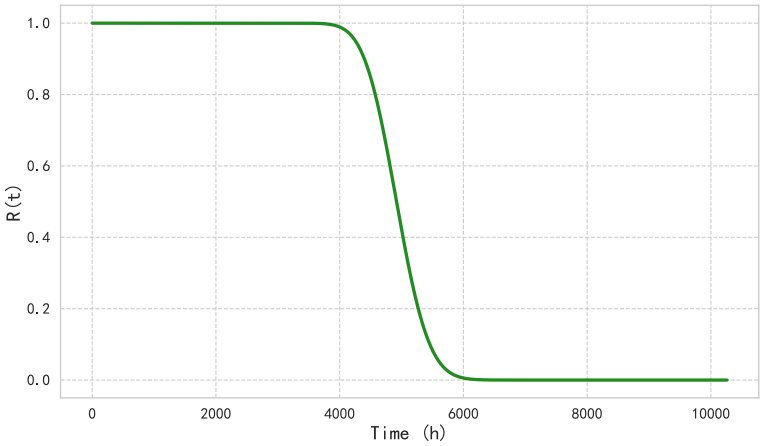

PDF of Failure Time



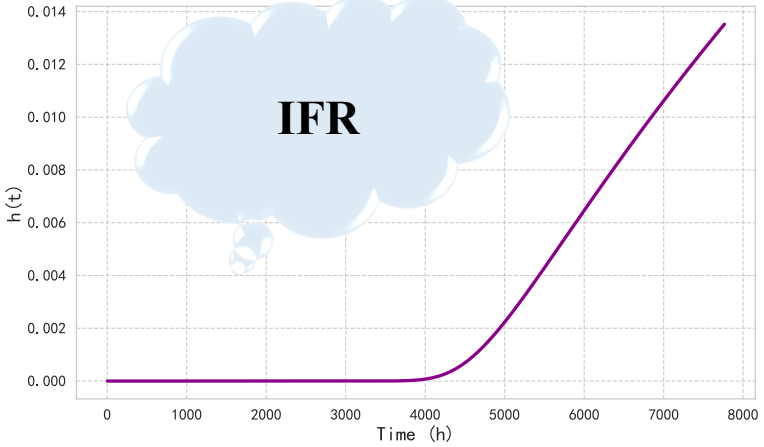

CDF of Failure Time




PDF of Degradation Increments

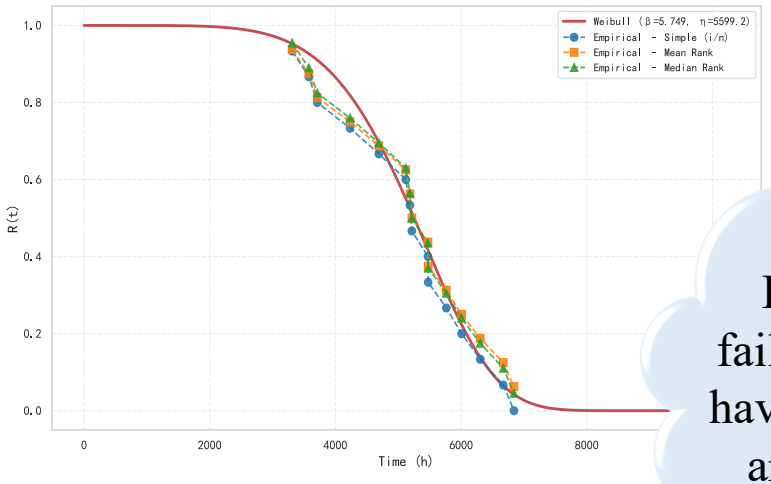



Reliability Function



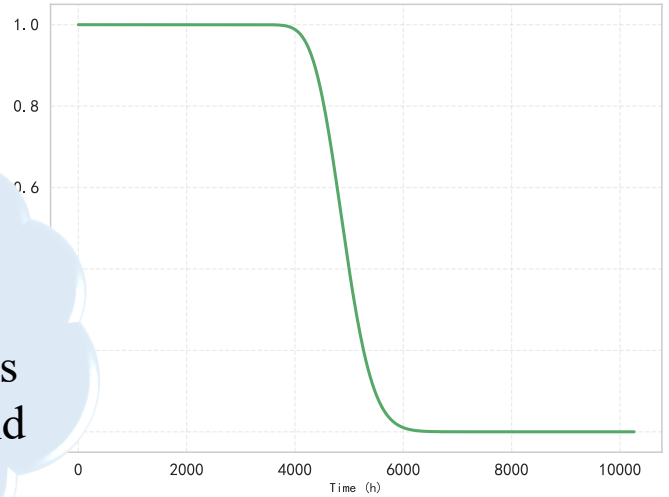

Failure Rate Function

# 基于随机过程的退化建模分析

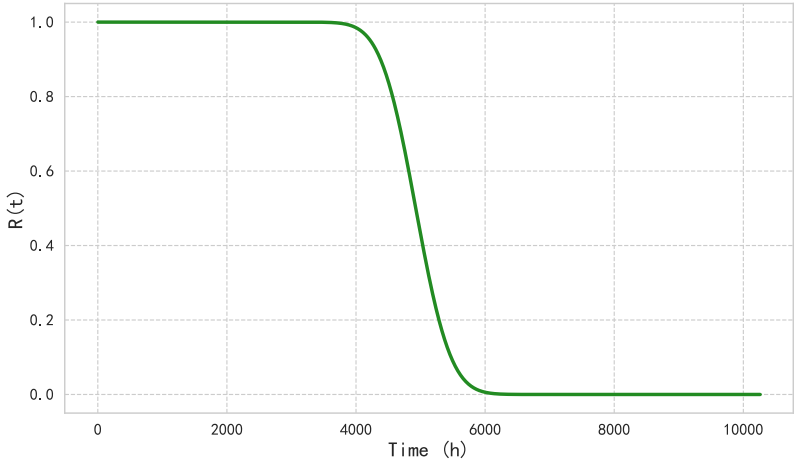



Failure Time Based Reliability

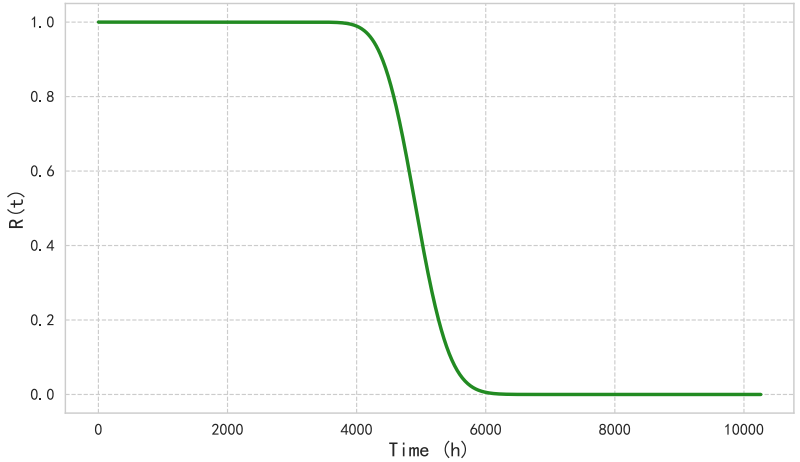
Degradation-based failure time estimations have lower variance and are more centralized




Wiener Degradation Process Based Reliability

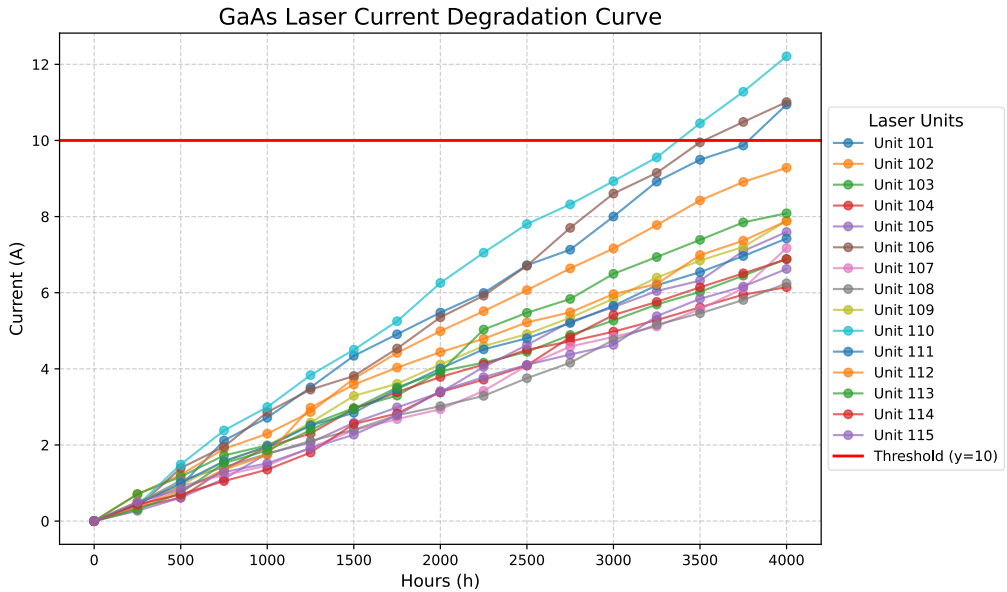


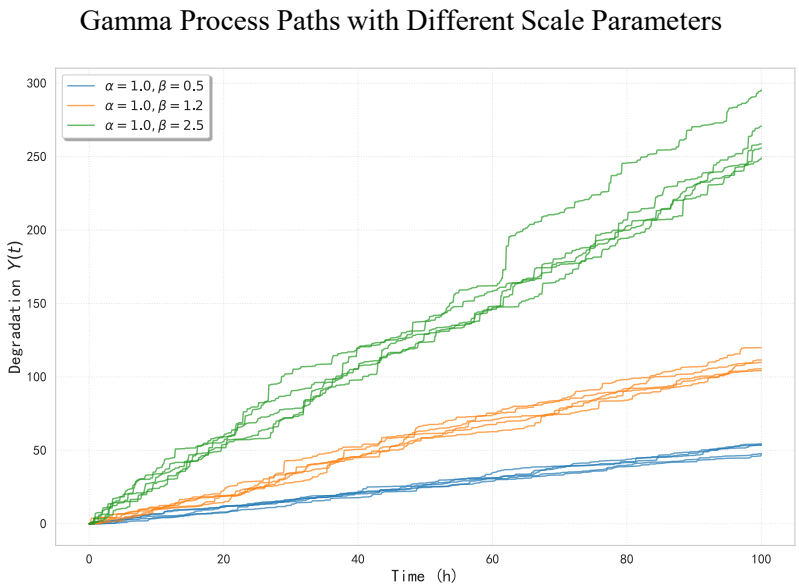

Inverse Gaussian Degradation Process Based Reliability




Gamma Degradation Process Based Reliability

# 基于随机效应Gamma过程的退化建模分析




**Unit-to-unit heterogeneity** is the inherent variability in degradation behavior across individual units of the same population. In stochastic modeling, it is captured via **Random Effects**, treating parameters (like drift or scale) as random variables to account for manufacturing tolerances and fluctuating environments.

# 基于随机效应Gamma过程的退化建模分析

Given $\beta_i$, the conditional increments of a path follow Gamma distribution: $\Delta y \mid \beta_i \sim \text{Gamma}(\alpha\Delta t, \beta_i)$

The conditional PDF of each increment is: $f(\Delta y \mid \alpha, \beta_i, \Delta t) = \dfrac{1}{\Gamma(\alpha\Delta t)\beta_i^{\alpha\Delta t}} \Delta y^{\alpha\Delta t - 1} \exp\left(-\dfrac{\Delta y}{\beta_i}\right)$

Furthermore, the scale para $\beta_i$ follows a Gamma distribution: $1/\beta_i \sim \text{Gamma}(\epsilon, \gamma)$

The prior PDF of latent variable $\beta_i$ is: $\pi(\beta_i \mid \epsilon, \gamma) = \dfrac{\gamma^{\epsilon}}{\Gamma(\epsilon)} \beta_i^{-(\epsilon+1)} \exp\left(-\dfrac{\gamma}{\beta_i}\right)$

**Latent Variable**

The marginal log-likelihood function is: $L_i = \int_0^{\infty} \left[ \prod_j \dfrac{\Delta y_{ij}^{\alpha\Delta t_{ij} - 1}}{\Gamma(\alpha\Delta t_{ij})\beta_i^{\alpha\Delta t_{ij}}} \exp\left(-\dfrac{\Delta y_{ij}}{\beta_i}\right) \right] \cdot \dfrac{\gamma^{\epsilon}}{\Gamma(\epsilon)} \beta_i^{-(\epsilon+1)} \exp\left(-\dfrac{\gamma}{\beta_i}\right) d\beta_i$

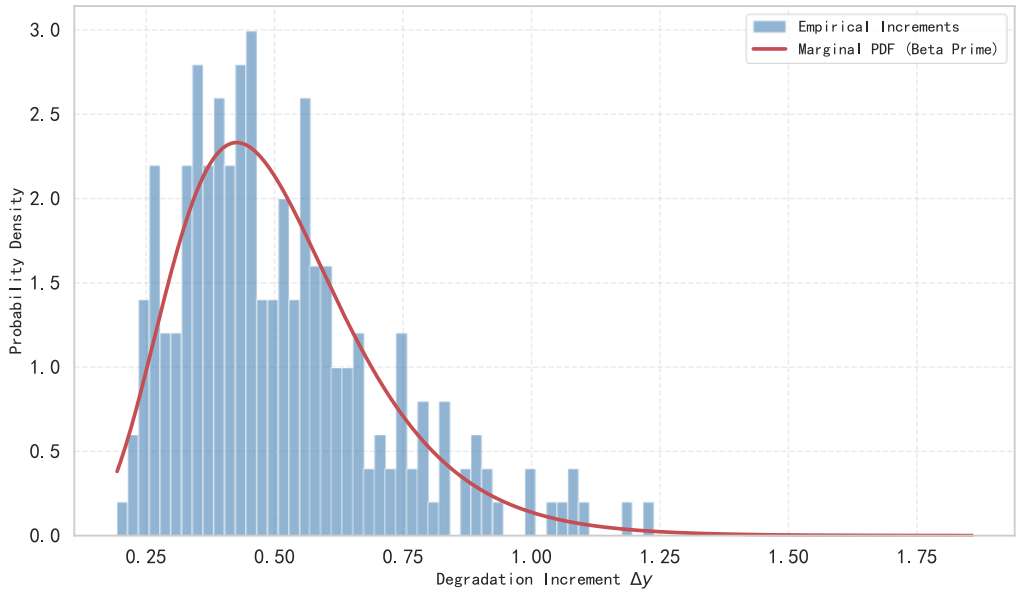
Group and merge all terms containing $\beta_i$: $L_i \propto \int_0^{\infty} \beta_i^{-(\alpha T_i + \epsilon + 1)} \exp\left(-\dfrac{Y_i + \gamma}{\beta_i}\right) d\beta_i$

Then the likelihood function can be obtained as: $L_i(\alpha, \epsilon, \gamma) = \dfrac{\Gamma(\alpha T_i + \epsilon)}{\Gamma(\epsilon)} \dfrac{\gamma^{\epsilon}}{(Y_i + \gamma)^{\alpha T_i + \epsilon}} \prod_j \dfrac{\Delta y_{ij}^{\alpha\Delta t_{ij} - 1}}{\Gamma(\alpha\Delta t_{ij})}$

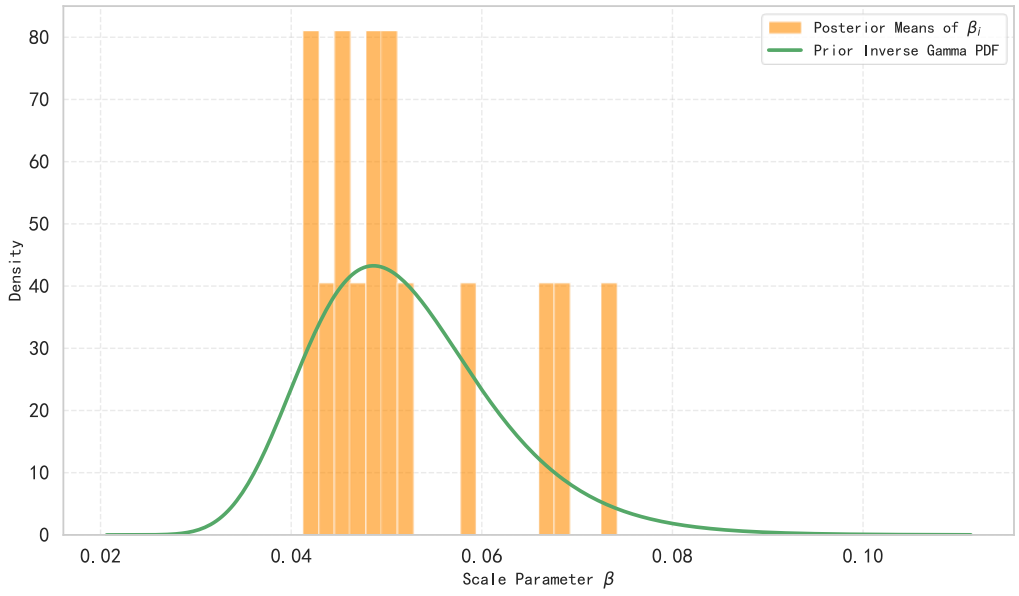
where: $T_i = \sum_{j=1}^{m} \Delta t_{i,j}, \quad Y_i = \sum_{j=1}^{m} \Delta y_{i,j}$

Parameters are estimated via MLE method

# 基于随机效应Gamma过程的退化建模分析




PDF of Increments with Histogram




Prior $\beta_i$ PDF with Posterior $\beta_i$ Histogram

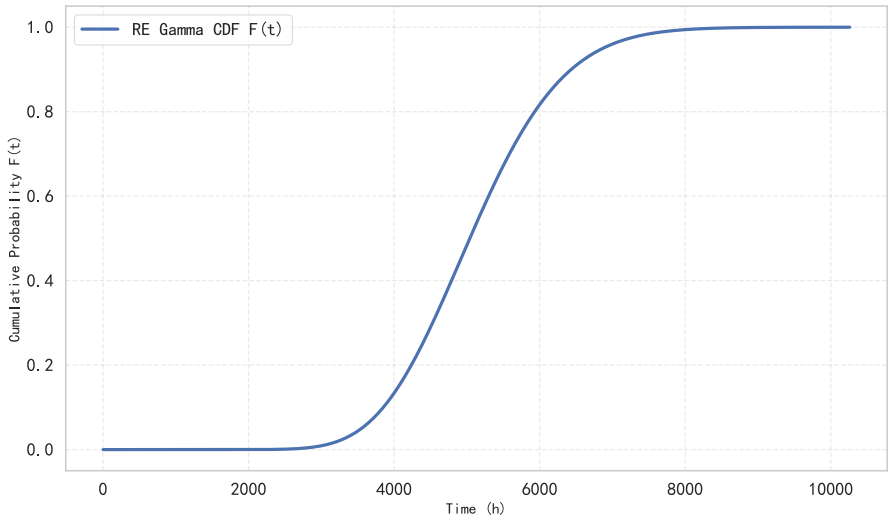
$$f(\Delta y_{i,j}) = \int_0^{+\infty} \frac{1}{\Gamma(\alpha\Delta t)\beta_i^{\alpha\Delta t}} \Delta y^{\alpha\Delta t-1} \exp\left(-\frac{\Delta y}{\beta_i}\right) \frac{\gamma^{\epsilon}}{\Gamma(\epsilon)} \beta_i^{-(\epsilon+1)} \exp\left(-\frac{\gamma}{\beta_i}\right) d\beta_i$$

$$= \frac{\Gamma(\alpha\Delta t_{i,j}+\epsilon)}{\Gamma(\epsilon)} \frac{\gamma^{\epsilon}}{(\Delta y_{i,j}+\gamma)^{\alpha\Delta t_{i,j}+\epsilon}} \frac{\Delta y_{ij}^{\alpha\Delta t_{ij}-1}}{\Gamma(\alpha\Delta t_{ij})}$$

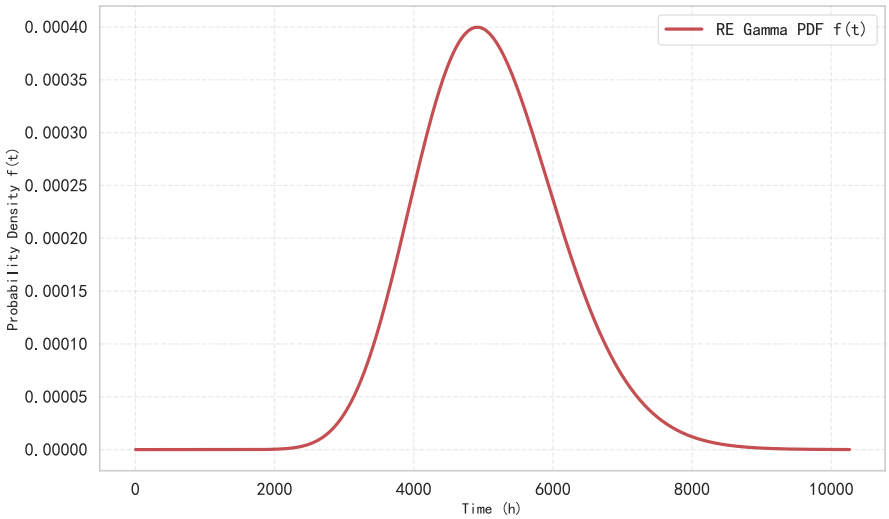
$$\beta_i \mid Y_i, T_i \sim \text{InverseGamma}(\epsilon+\alpha T_i, \gamma+Y_i)$$

$$\alpha = 3.905973e-02, \quad \epsilon = 2.892961e+01, \quad \gamma = 1.455415e+00$$

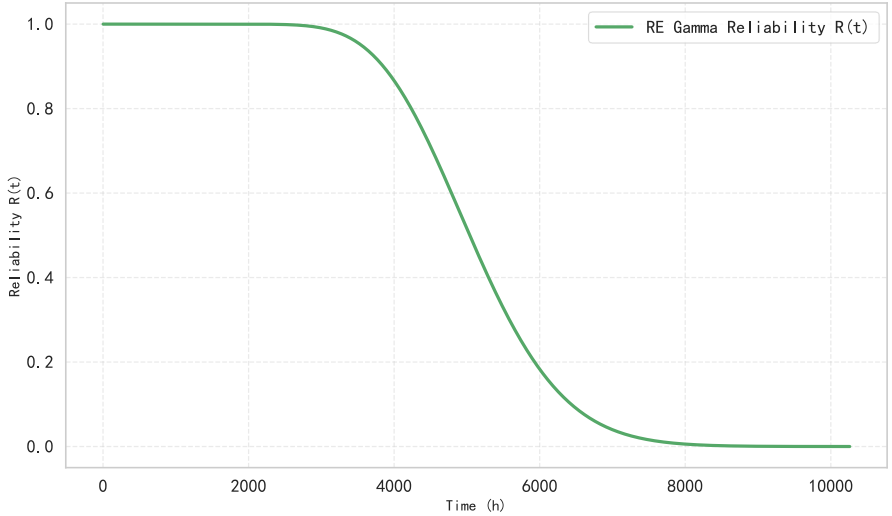
# 基于随机效应Gamma过程的退化建模分析



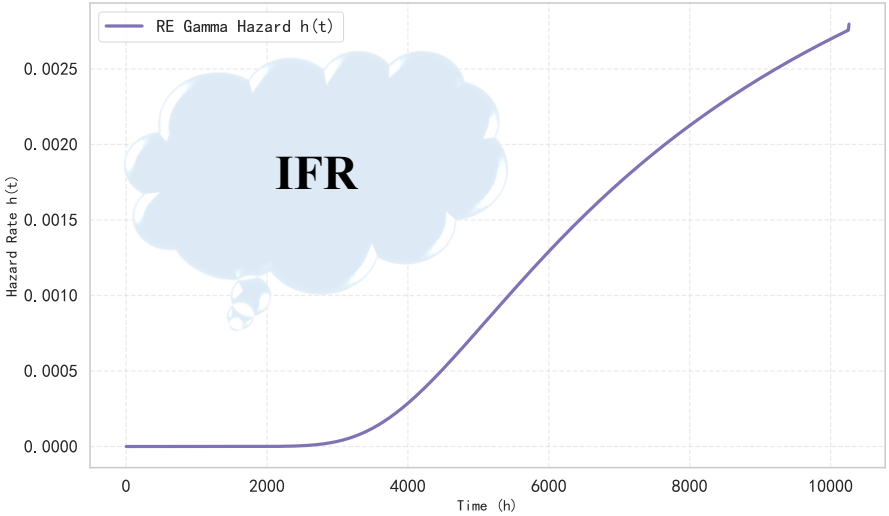

CDF of Failure Time




PDF of Failure Time




Reliability Function




Failure Rate Function

# 基于随机效应Gamma过程的退化建模分析

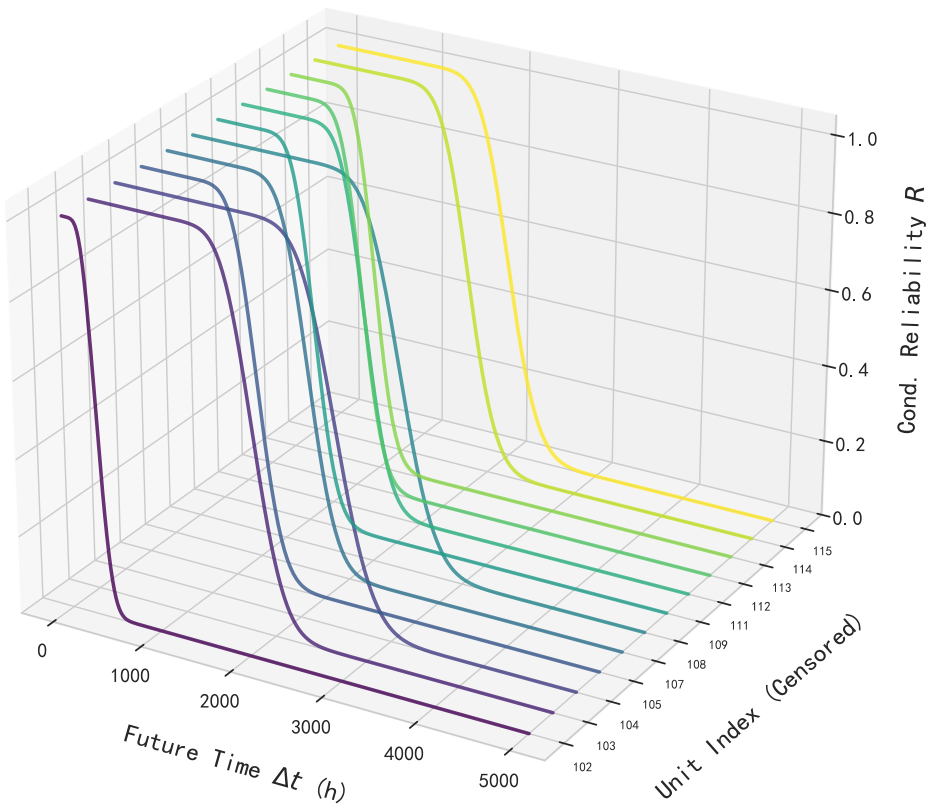
$$R(s\mid Y_i,T_i)=\int_0^{\infty}P(\Delta X(s)<\delta-Y_i\mid\beta_i)\cdot\pi(\beta_i\mid Y_i,T_i)d\beta_i$$

$$=\frac{\Gamma(\alpha s+\alpha T_i+\epsilon)}{\Gamma(\alpha s)\Gamma(\alpha T_i+\epsilon)}(Y_i+\gamma)^{\alpha T_i+\epsilon}\int_0^{\delta-Y_i}\frac{x^{\alpha s-1}}{(x+Y_i+\gamma)^{\alpha s+\alpha T_i+\epsilon}}dx$$

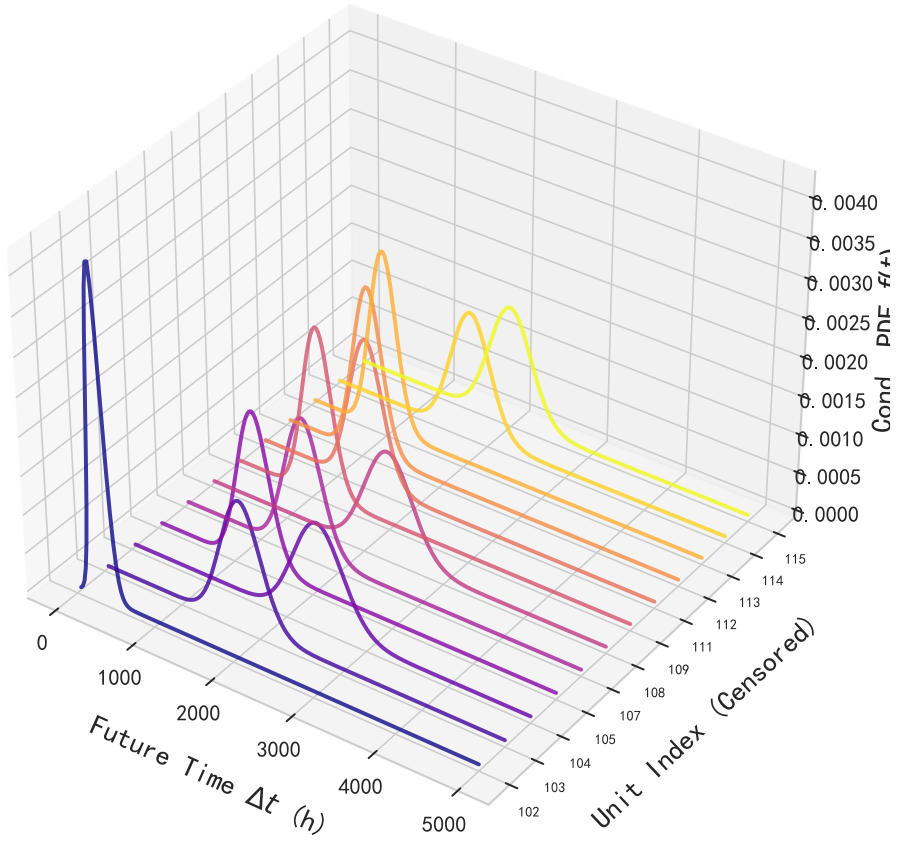
$$f(s\mid Y_i,T_i)=\frac{\partial R(s\mid Y_i,T_i)}{\partial s}$$

$$=\frac{\partial}{\partial s}\left[\frac{\Gamma(\alpha s+\alpha T_i+\epsilon)}{\Gamma(\alpha s)\Gamma(\alpha T_i+\epsilon)}(Y_i+\gamma)^{\alpha T_i+\epsilon}\int_0^{\delta-Y_i}\frac{x^{\alpha s-1}}{(x+Y_i+\gamma)^{\alpha s+\alpha T_i+\epsilon}}dx\right]$$

Monte Carlo Method

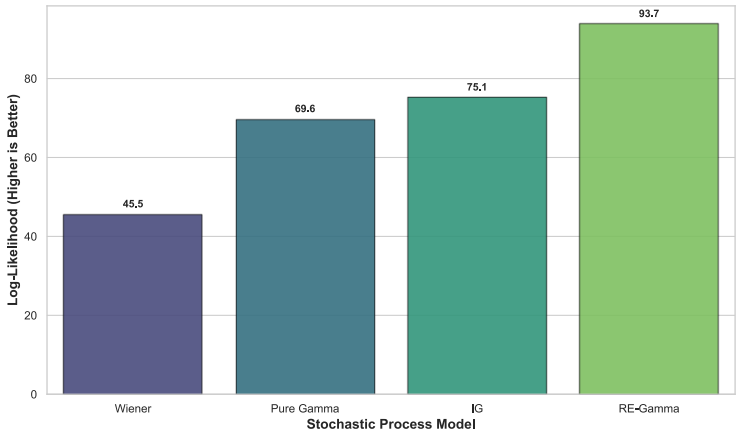



Conditional Reliability

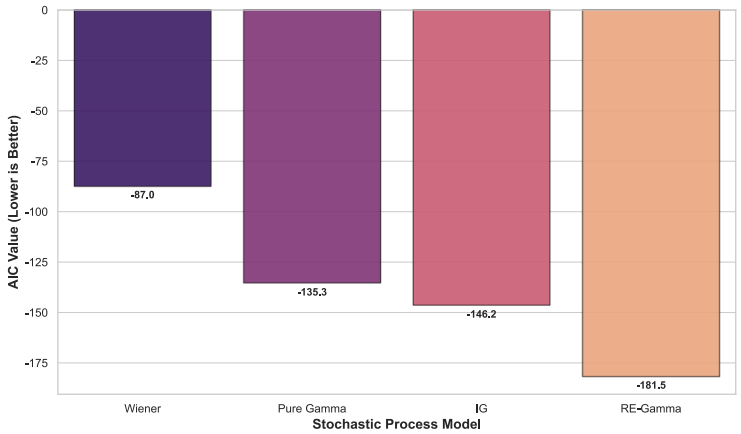



Conditional PDF of Future Time

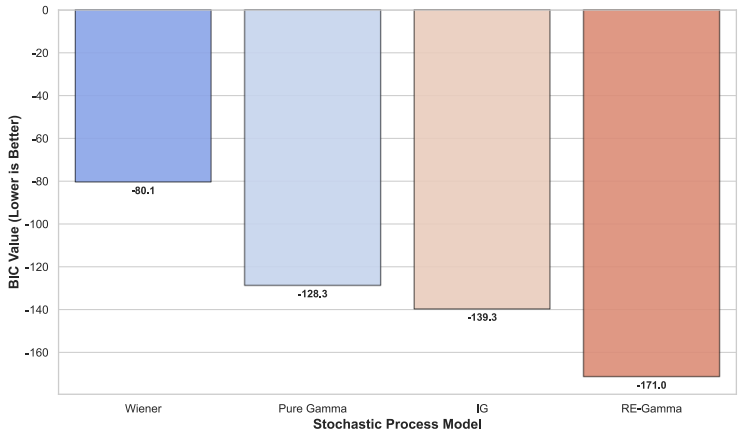
# 退化模型对比




Log-likelihood Value




AIC Value




BIC Value

$$\ln\left[L(\theta)\right] = \sum_{i=1}^{N} \ln\left[f(\theta; x_i)\right]$$

$$AIC = -2\ln(L) + 2k$$

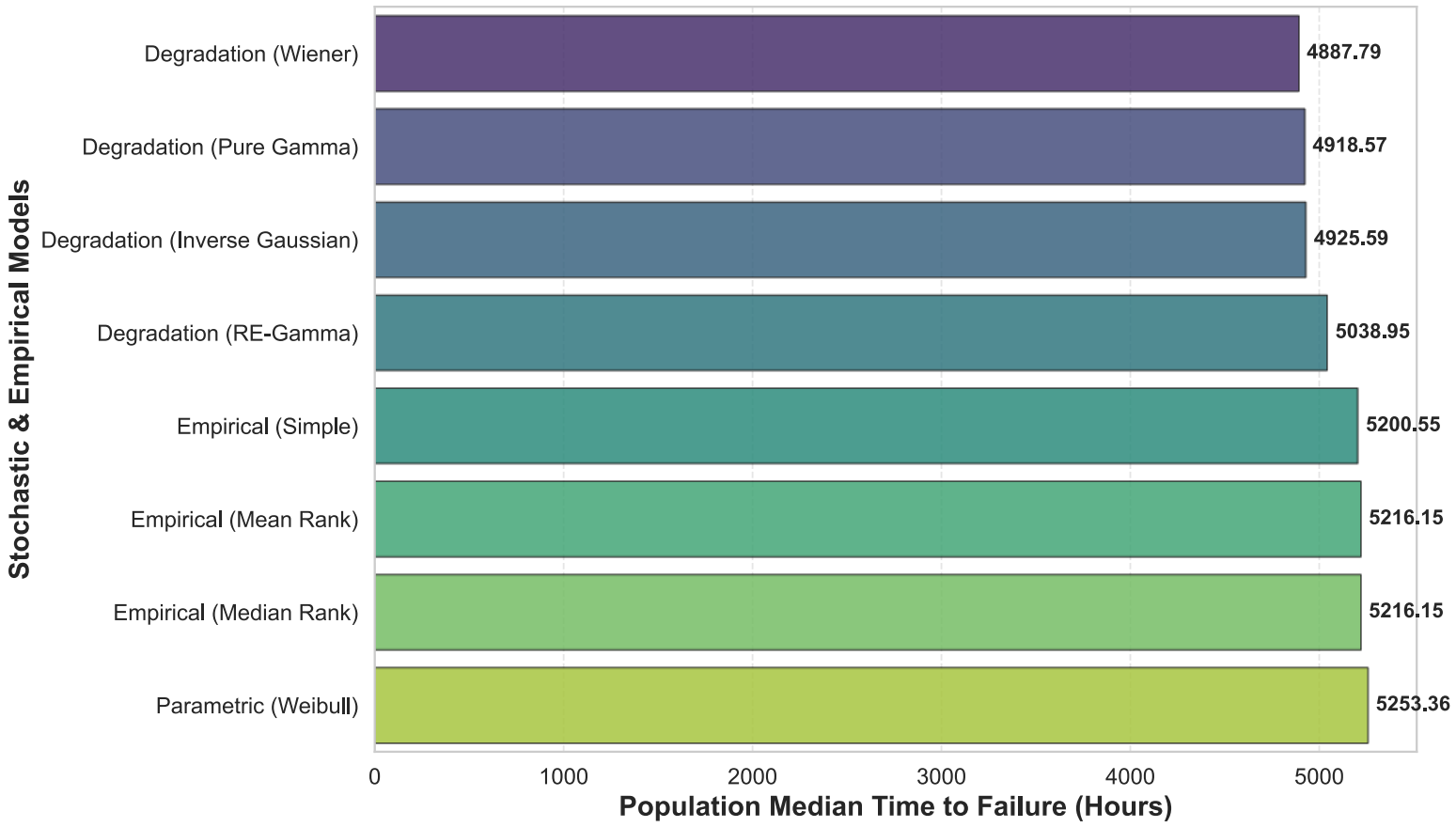
$$BIC = -2\ln(L) + k\ln(n)$$

Log-Likelihood measures the model's **goodness-of-fit** by quantifying how well the estimated parameters explain the observed data; a **higher** value indicates a more probable fit

AIC evaluates the trade-off between model fit and **complexity** by penalizing the number of parameters to minimize information loss; a **lower** value indicates a more parsimonious and efficient model

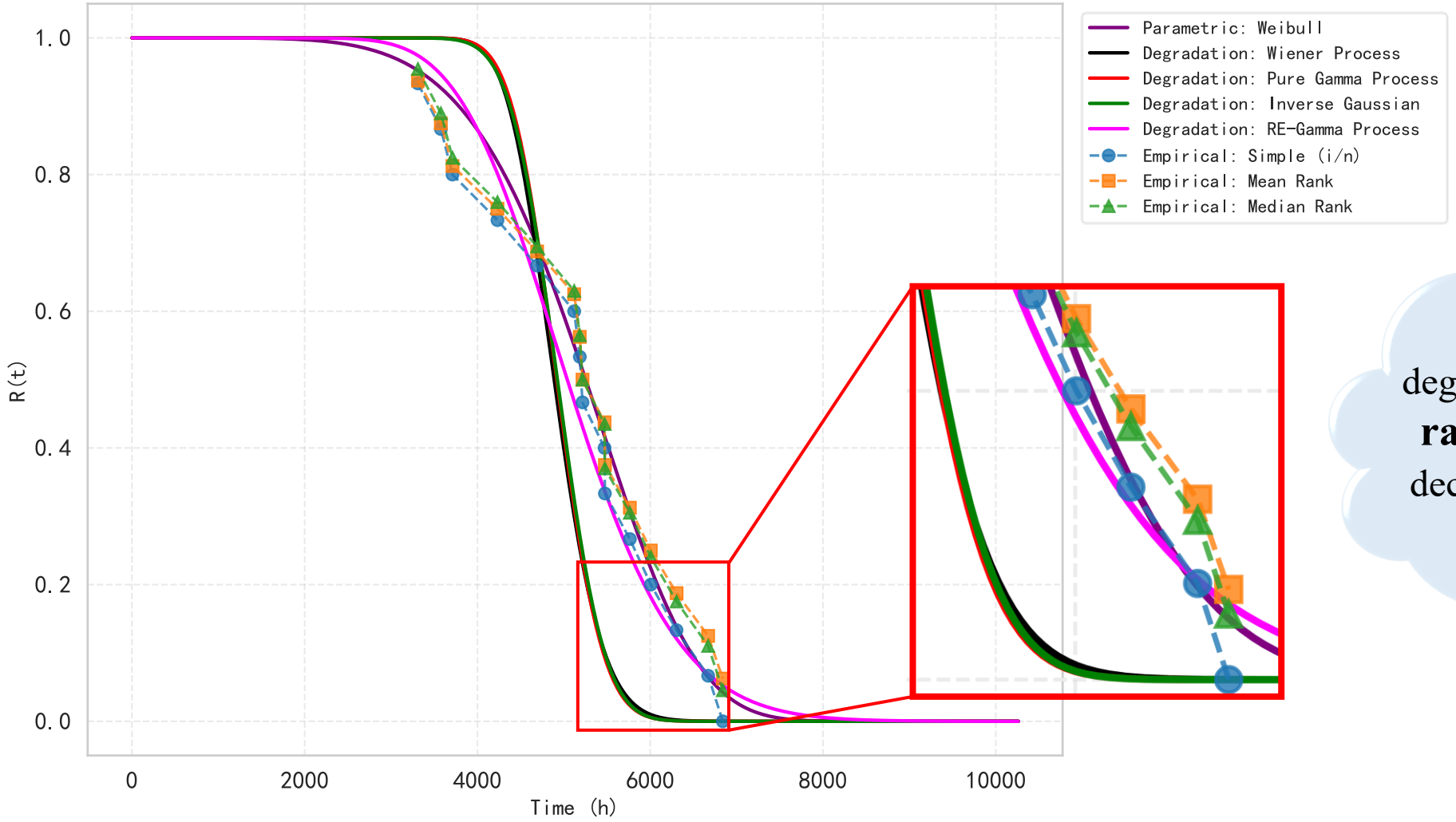
BIC balances fit against a stricter complexity penalty that scales with **sample size**, favoring simpler models to identify the "true" underlying process; a **lower** value is preferred

# 模型对比——中位失效时间




Median Time to Failure across Models

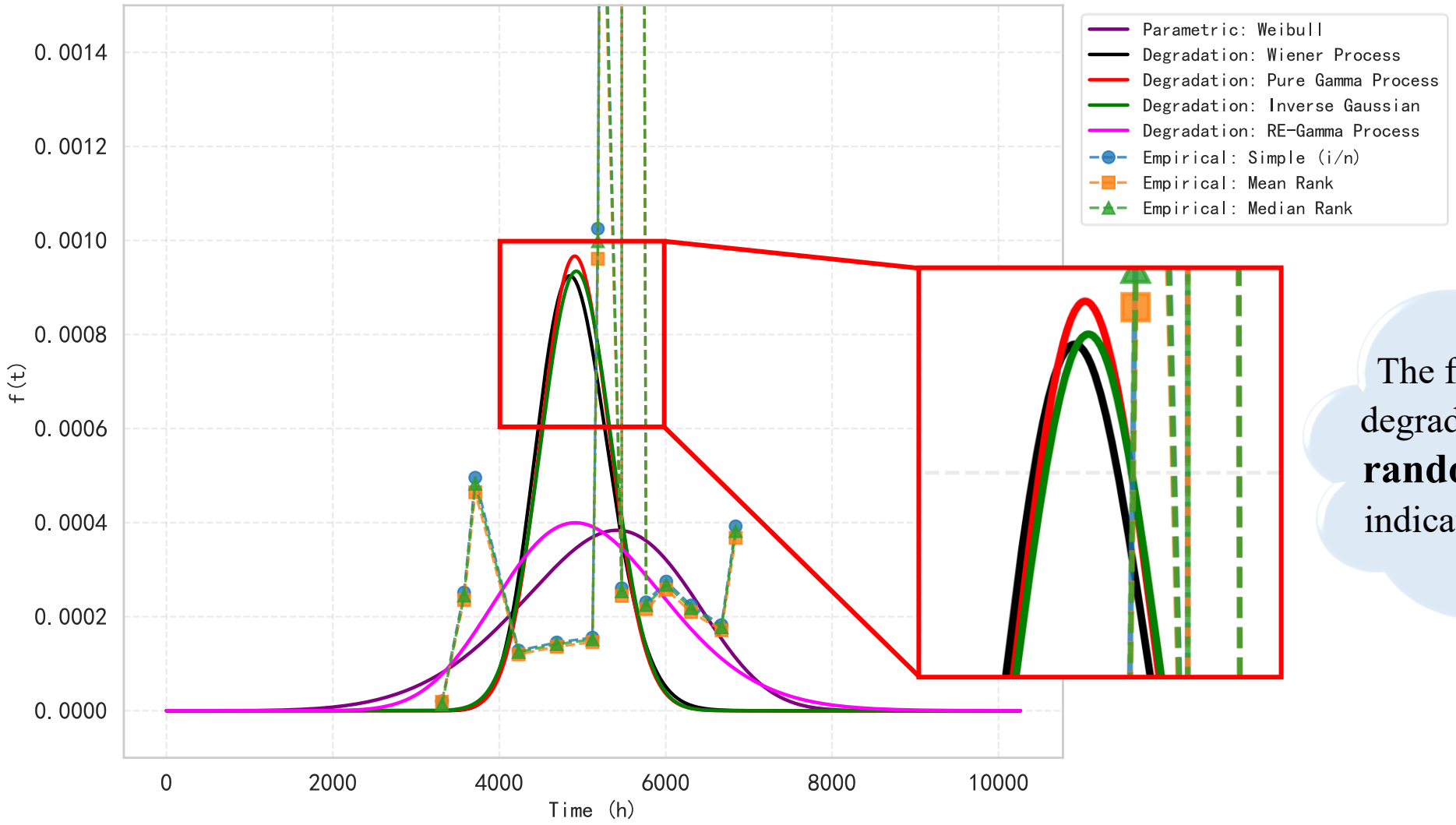
# 模型对比——可靠度函数




Reliability Curve across Models

The reliability of the degradation model **without random effects** starts to decline **later** but decreases more **rapidly**

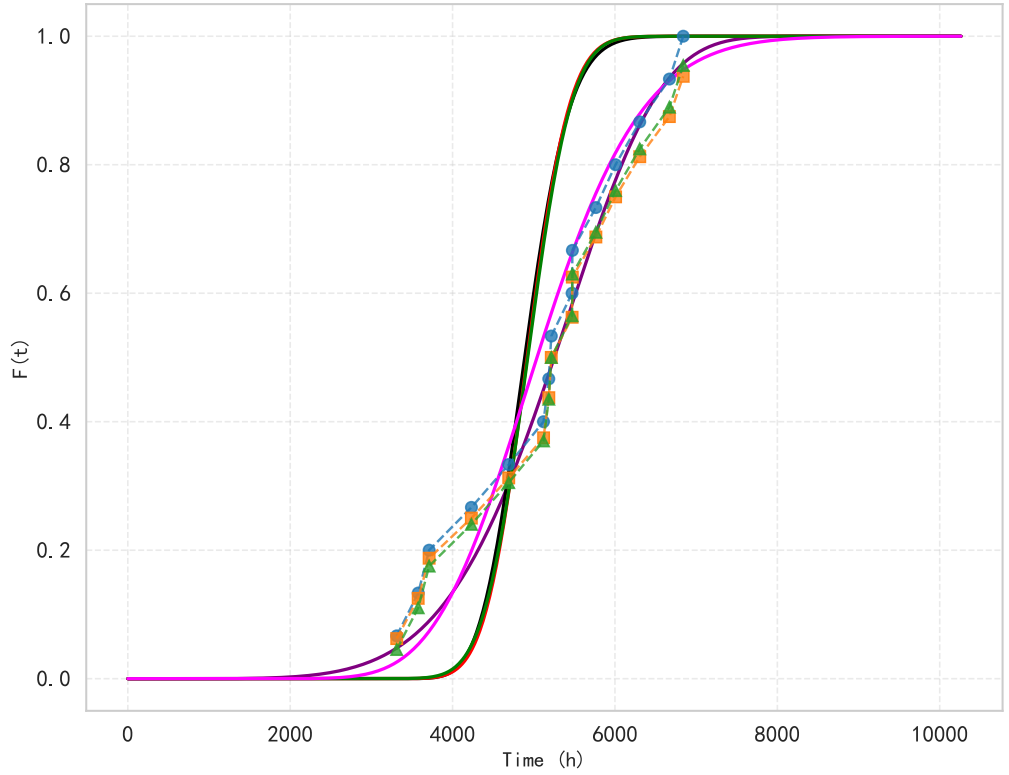
# 模型对比——失效时间PDF



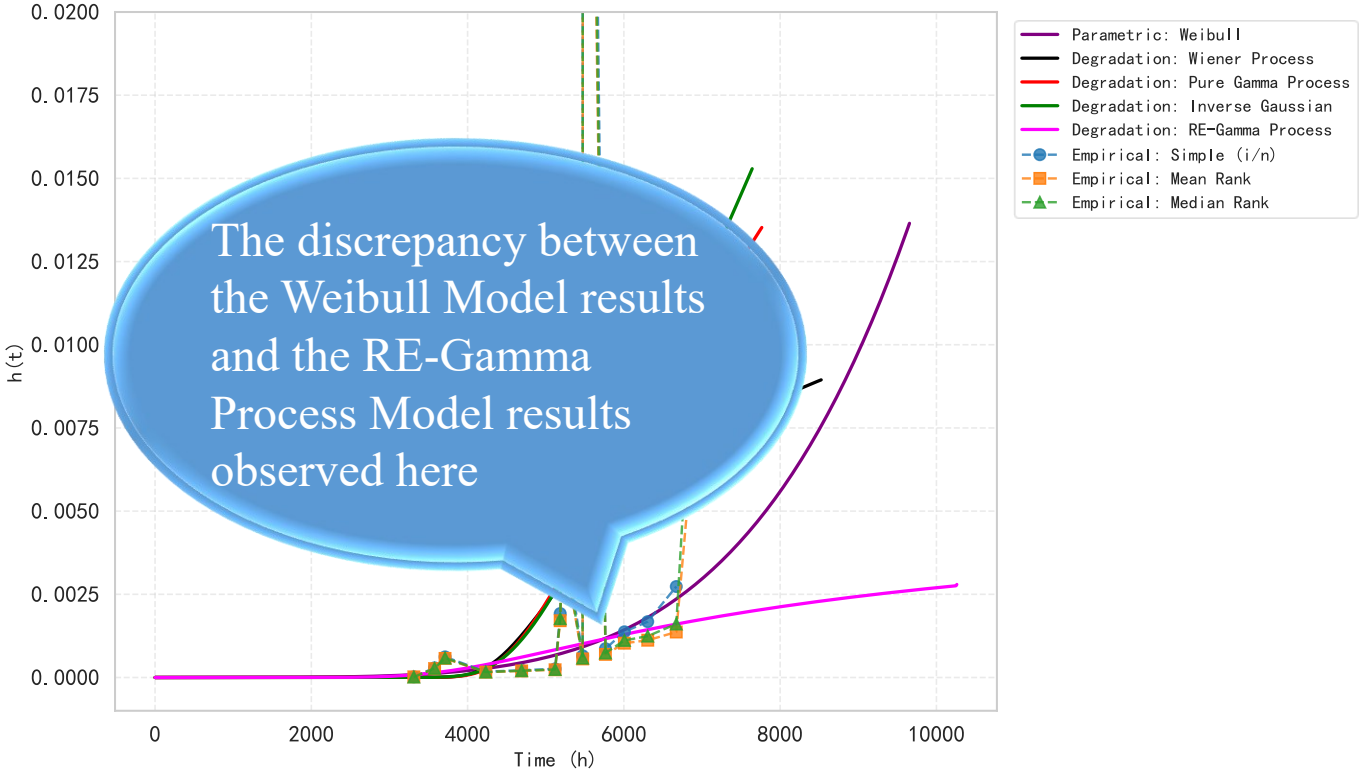

PDF of Failure Time across Models

The failure time PDF for the degradation model **excluding random effects** is **sharper**, indicating **lower variability**

# 模型对比——CDF与失效率

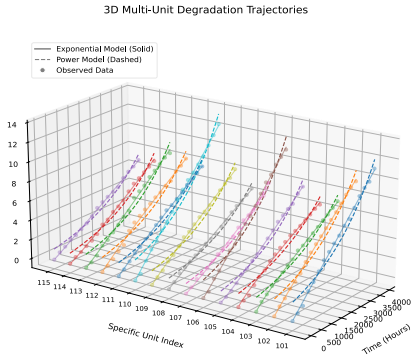



CDF of Failure Time across Models

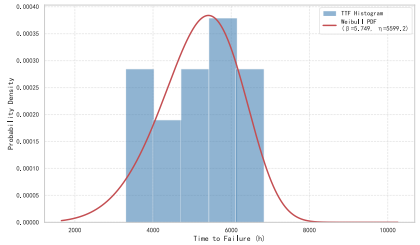



Failure Rate Function across Models

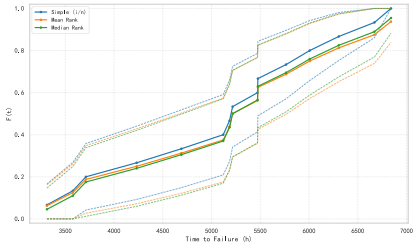
# Overall Project Review



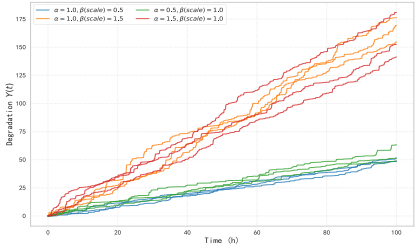

Degradation Spline Fit



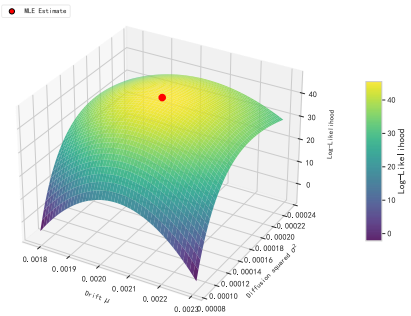

Weibull Distribution Fit and Analysis



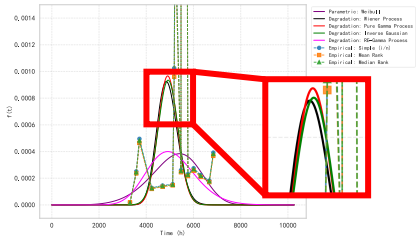

Empirical Method Assessment



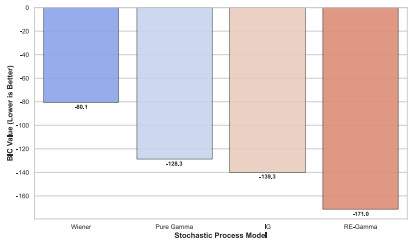

Degradation Process based Analysis



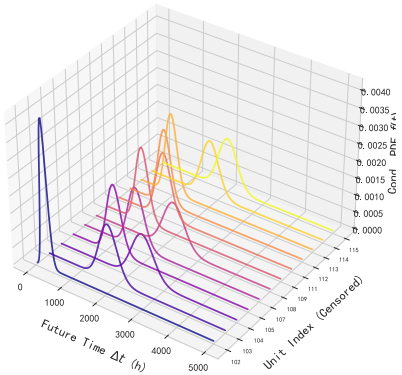

MLE Parameter Estimation Method



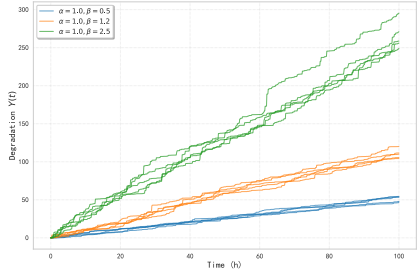

Overall Model Comparison



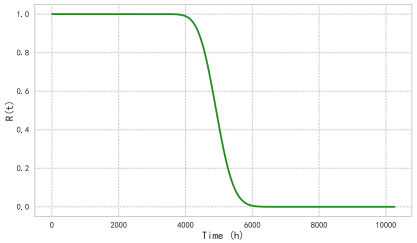

Degradation Model Selection




Conditional Reliability Assessment




Random Effects Degradation Model




Degradation based Assessment

GitHub [Term6Assignments/Reliability/project1/data/课题一 at main · Promileee/Term6Assignments]

# Thanks

电子科技大学
University of Electronic Science and Technology of China